

Reconstruction of long-lived particles with the CLUE algorithm in CMS

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Abstract

Long-lived particles can decay far from the beamline, producing photons whose trajectories do not point back to the interaction point. This breaks the assumption behind CLUE3D, the CMS clustering algorithm for the High-Granularity Calorimeter, which groups energy along directions projected from the origin. Using controlled single- and two-photon simulations, with and without pileup, I study how well such displaced photons are reconstructed as a function of their transverse displacement.

I first explain geometrically why default CLUE3D loses displaced photons: its origin projection makes points on a single inclined shower appear transversely separated, so the shower fragments as displacement grows, while in two-photon events nearby showers merge. The main result is that alternative CLUEstering metrics can do better, recovering displaced photons more completely while keeping neighbouring showers apart. Tuning of the coordinates, distance metrics and density settings is ongoing, but the results already point to a clustering better suited to non-pointing photons than the origin-projected default.

1 Introduction

The year 2031 will mark the start of the High-Luminosity phase of the Large Hadron Collider (HL-LHC) [1], which will deliver proton–proton collisions at an unprecedented instantaneous luminosity. The resulting data volume and the severe pileup—of order 200 simultaneous interactions per bunch crossing—demand a new generation of reconstruction algorithms. To cope with these conditions the CMS experiment will replace its endcap calorimeters with the High-Granularity Calorimeter (HGCAL) [2], a sampling calorimeter with roughly six million channels arranged in about fifty longitudinal layers. Its fine transverse and longitudinal segmentation records the detailed three-dimensional structure of electromagnetic and hadronic showers, but it also makes traditional calorimeter clustering impractical and motivates dedicated pattern-recognition algorithms.

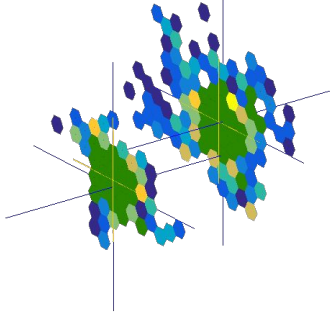
The Clustering of Energy (CLUE) algorithm [3] was developed for exactly this purpose: it is a fast, parallel, density-based clustering algorithm that groups HGCAL energy deposits into physics objects such as photons and electrons. In the CMS reconstruction, CLUE and its three-dimensional extension CLUE3D sit inside The Iterative Clustering (TICL) framework [4], which builds objects in stages (Fig. 1): individual calorimeter hits (*RecHits*) are first grouped by CLUE into per-layer two-dimensional *LayerClusters* (LCs); CLUE3D then links *LayerClusters* across layers into three-dimensional *Tracksters*; and a subsequent linking step associates *Tracksters* into the final TICL candidates. The Monte Carlo truth is described by an analogous chain, in which a generated particle (*CaloParticle*) is mapped to a *SimTrackster* that collects the *LayerClusters* truly belonging to it. Reconstruction performance is quan-

tified by comparing reconstructed *Tracksters* to *SimTracksters* through their shared energy, as detailed below.

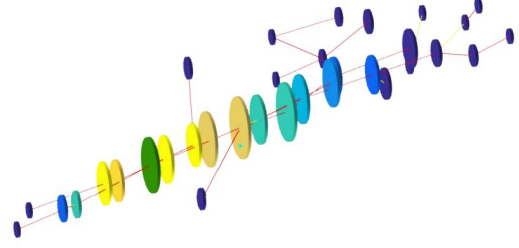
A key assumption underlies CLUE3D: the neighbour search that links layer clusters across layers is performed in a coordinate system *projected from the interaction point*, so that a shower travelling radially outward from the detector centre keeps a fixed transverse position layer to layer. This holds for prompt particles produced at the beamline, but it doesn't need to for the class of *long-lived particles* (LLPs) predicted by many extensions of the Standard Model [5]. Rather than decaying within a few millimetres of the collision point, an LLP can traverse a sizeable fraction of the detector before decaying, producing *displaced*, non-pointing particles. A representative signature is a hidden-valley dark shower, in which long-lived dark hadrons decay to Standard Model photons far from the beamline. Because these showers do not point back to the origin, they may violate exactly the assumption on which CLUE3D relies.

The goal of this project is to determine whether CLUE is capable of clustering the energy deposits arising from such novel topologies, to quantify the expected performance reduction as a function of the distance from the detector centre at which the LLP decays, and to explore an alternative algorithm. We restrict the study to neutral *photons*: photons are not deflected by the CMS magnetic field, so their trajectories are straight lines and their displacement and incidence angle are unambiguously defined; they also do not interact in the tracker before reaching the calorimeter. This isolates the calorimeter clustering behaviour from tracking and magnetic-field effects and provides the cleanest possible test of the origin assumption.

The remainder of this report is organized as follows.



(a) RecHits and LayerClusters



(b) LayerClusters linked into a Trackster

Figure 1: The HGCAL reconstruction chain. (a) Calibrated energy deposits (RecHits) of two photon showers, which CLUE groups within each layer into two-dimensional LayerClusters (coloured by deposited energy). (b) CLUE3D links LayerClusters (disks) across layers into a three-dimensional Trackster; the red segments are the nearest-higher links that build the shower.

We first define the displacement observables and describe how they are reconstructed for objects that have no associated truth vertex, then present the displaced-photon Monte Carlo production and the performance metrics. After reviewing how CLUE3D works, with emphasis on the origin assumption, we present the single-photon efficiency, the two-shower merging behaviour, and the performance under pileup, before comparing CLUE3D against the alternative algorithm CLUEstoring and concluding.

2 Displacement observables

Every result in this report is expressed as a function of *how displaced* a shower is. We therefore begin by defining the displacement observables and, crucially, by explaining how they are obtained for reconstructed objects that carry no truth information.

2.1 Definitions

Consider a photon that is created at a production vertex $\mathbf{x}_v = (x_v, y_v, z_v)$ and travels in a straight line to the HGCAL front face, located at $z \simeq 321$ cm from the interaction point (Fig. 2). We characterize its displacement by two quantities:

- the **displacement radius** R , defined as the transverse distance from the beamline at which the photon's extrapolated trajectory crosses the plane $z = 0$,

$$R = \sqrt{x_0^2 + y_0^2}, \quad (1)$$

where (x_0, y_0) is the trajectory back-projected to $z = 0$;

- the **non-pointing angle** α , the angle between the photon's momentum $\mathbf{p}$ and its position vector at the HGCAL face, i.e. the line joining the interaction point to the impact point $\mathbf{x}_{\text{face}}$,

$$\alpha = \arccos\left(\frac{\mathbf{x}_{\text{face}} \cdot \mathbf{p}}{|\mathbf{x}_{\text{face}}| |\mathbf{p}|}\right). \quad (2)$$

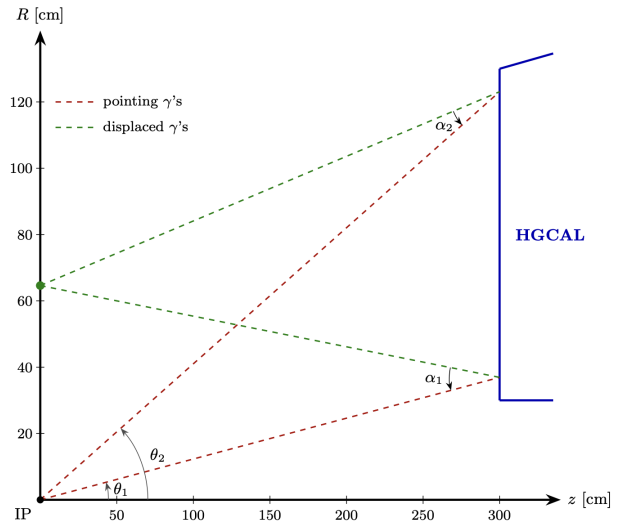


Figure 2: Displacement geometry in the (z, R) plane. Pointing photons (red) travel radially from the interaction point (IP) and reach the HGCAL front face at a polar angle θ with non-pointing angle $\alpha = 0$. Displaced photons (green) originate at a transverse distance R from the beamline at $z = 0$ and strike the calorimeter with $\alpha > 0$; two displaced trajectories from the same vertex illustrate that a single R can correspond to different incidence angles α .

A prompt, pointing photon travels radially, so $\mathbf{x}_{\text{hit}} \parallel \mathbf{p}$ and $R = \alpha = 0$; a displaced photon has $R > 0$ and $\alpha > 0$.

It is also useful to introduce the polar angle θ between the photon momentum and the beam axis, with $\tan \theta = p_T/|p_z|$. As shown in Fig. 2, α rather than R is the more fundamental driver of the clustering behaviour, but R has a direct geometric interpretation as the transverse distance, so both are retained throughout.

For a *simulated* photon these quantities are trivial: the production vertex and momentum are known from the generator, and R follows from a straight-line extrapolation. We verified this extrapolation numerically—the displacement computed from the SimTrackster's SimTrack reproduces the generator-level value, and pointing photons correctly yield $R, \alpha \approx$

0.

2.2 Reconstructing the displacement

The difficulty lies on the reconstruction side. A reconstructed Trackster has no production vertex, and it does not need to correspond one-to-one to a SimTrackster: it may contain only a fragment of a true shower, or it may span multiple showers. Nonetheless, the fake-, duplicate- and merge-rate metrics defined below require a displacement value in order to place each Trackster in a histogram bin. We therefore reconstruct R directly from the Trackster geometry.

Each Trackster provides an energy-weighted barycentre $\mathbf{x}_c = (x_c, y_c, z_c)$ and a principal-axis direction $\mathbf{u} = (u_x, u_y, u_z)$ obtained from a principal-component analysis (PCA) of its LayerClusters. Treating the shower as a straight line through $\mathbf{x}_c$ along $\mathbf{u}$ and extrapolating to $z = 0$ gives

$$x_0 = x_c - z_c \frac{u_x}{u_z}, \quad y_0 = y_c - z_c \frac{u_y}{u_z}, \quad (3)$$

and $R_{\text{reco}} = \sqrt{x_0^2 + y_0^2}$ from Eq. (1). Care is taken so that the extrapolation uses the correct sign of u_z . This straight-line treatment is exact for photons, which makes them the ideal probe for validating the method.

2.3 Error budget and the choice of direction estimator

The accuracy of R_{reco} sets the resolution with which any reco-side metric can be studied, so we characterized its error in detail. We quantify it by

$$\Delta R \equiv R_{\text{reco}} - R_{\text{truth}}, \quad (4)$$

summarized throughout by its RMS. Our starting point was the direction estimator that CMSSW already exposes: the ordinary PCA of `TrackstersPCA.cc`, which builds the energy-weighted covariance matrix of the LayerCluster positions about the barycentre, and makes the shower direction $\mathbf{u}$ to be its leading eigenvector (the axis of largest spatial variance). Reusing this default out of the box gave a broad error— $\text{RMS}(\Delta R) \approx 20$ cm for the sample of Fig. 3—and before trying to improve it we wanted to know *where* that error came from.

The error has only two possible sources, the barycentre $\mathbf{x}_c$ and the direction $\mathbf{u}$, and they can be separated cleanly. We recompute ΔR twice: once with the reconstructed barycentre but the *true* direction (isolating the barycentre error), and once with the true barycentre but the reconstructed direction (isolating the direction error). The answer is unambiguous (Fig. 3): the barycentre alone contributes only $\text{RMS}(\Delta R) = 1.14$ cm, whereas the direction alone contributes 20.60 cm and by itself reproduces the 20.46 cm of the full error. The barycentre is essentially perfect; the direction is entirely to blame. The reason is geometric: the barycentre sits inside the shower and is well measured, but the small angular uncertainty of $\mathbf{u}$

is amplified by the long $z_c \sim 320$ cm over which Eq. (3) extrapolates back to $z = 0$. The same figure shows a systematic positive bias of order +3 cm, consistent with R being a positive-definite quantity.

Having localized the problem to the direction, we implemented and compared several estimators of the shower axis. Each takes the LayerClusters (or RecHits) of a Trackster and returns a unit direction:

- **Ordinary PCA** (the CMSSW default): leading eigenvector of the energy-weighted covariance matrix $C = \sum_i w_i (\mathbf{x}_i - \mathbf{x}_c)(\mathbf{x}_i - \mathbf{x}_c)^\top$, where $\mathbf{x}_i$ and w_i are the position and energy of LayerCluster i . It uses every LayerCluster, so secondary per-layer clusters and stray fragments can tilt the axis.
- **Clean PCA (10-to-10)**: the same eigenvector computation, but on a cleaned set that keeps only the single highest-energy LayerCluster in each layer, restricted to the ten layers before and the ten layers after the peak-energy layer. This suppresses the off-axis fragments that bias the covariance while retaining the well-measured shower core.
- **Endpoint layers**: the direction joining the barycentre of the first populated layer of the window to that of the last. It uses only the two extremes, so it is simple but sensitive to endpoint fluctuations.
- **Window barycentres**: per-layer energy-weighted barycentres are formed across the window and their trend in z defines the axis, averaging out single-layer noise.
- **Weighted straight-line fit**: the shower is parametrized as a line $x(z) = x_0 + (u_x/u_z)z$, $y(z) = y_0 + (u_y/u_z)z$, and the slopes are obtained by minimizing the energy-weighted sum of squared transverse residuals of the LayerClusters to that line—a direct least-squares alternative to the PCA.
- **Weighted line fit with outlier refit**: the same fit, after which the 20% of LayerClusters with the largest residuals are discarded and the line is refitted, reducing the pull of outliers; we tried this both on the 10-to-10 window and over all layers.
- **RecHit PCA**: the covariance-eigenvector method applied to the raw RecHits instead of the LayerClusters, testing whether the finer granularity below the LayerCluster level helps.

The comparison (Fig. 4) makes clear that no estimator improves decisively on the default. On the leading Trackster (most energetic one per event, closest to the truth) the PCA variants are already the best and the more elaborate line fits merely match them; over all Tracksters (which includes small fragments) every method collapses to the same ~ 20 cm, so the estimator is almost irrelevant and it is the Trackster selection that drives the resolution. Because the clean PCA is

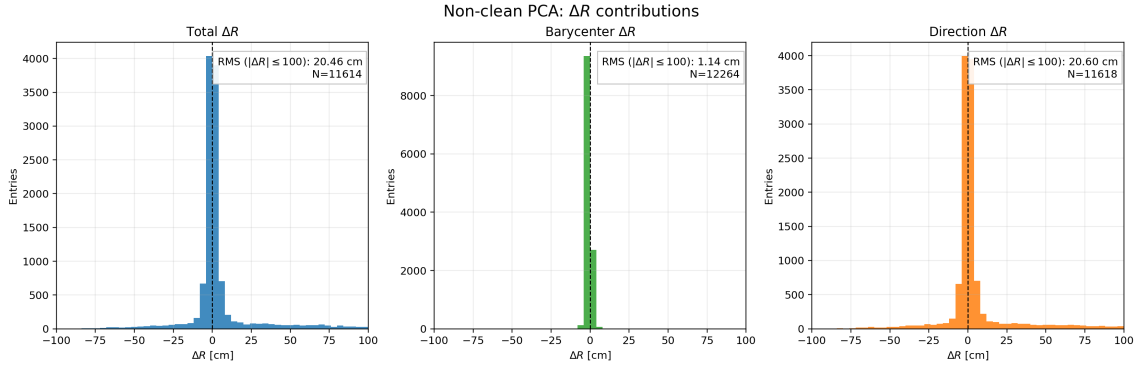


Figure 3: Decomposition of the displacement-radius error into its two sources, for the default (ordinary) PCA. *Left*: the full error $\Delta R = R_{\text{reco}} - R_{\text{truth}}$ (RMS = 20.46 cm). *Centre*: the error when only the barycentre is reconstructed and the true direction is used (RMS = 1.14 cm). *Right*: the error when only the direction is reconstructed and the true barycentre is used (RMS = 20.60 cm). The direction term alone reproduces the full error, proving that the resolution budget is spent almost entirely on the shower direction.

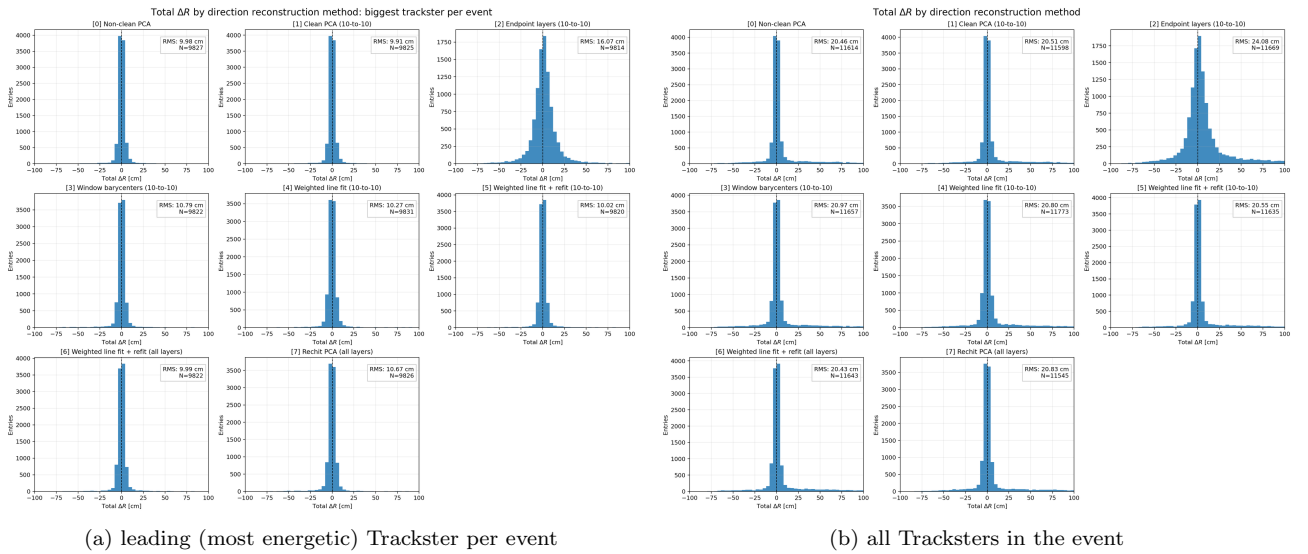


Figure 4: Full-error distribution ΔR for each of the eight direction estimators, (a) restricting to the leading Trackster and (b) over all Tracksters. Restricting to the leading Trackster, the PCA variants are the narrowest (clean PCA 9.91 cm, ordinary 9.98 cm) and the endpoint estimate the broadest (16.07 cm). Over all Tracksters every estimator instead collapses to ~ 20 cm. The choice of estimator barely matters, because the sample is then dominated by low-energy fragments whose direction no method can fit.

both robust and already implemented in CMSSW, it was adopted as the default.

This picture is confirmed by binning the resolution in Trackster observables (Fig. 5): the RMS falls from ~ 48 cm below 10 GeV to ~ 10 cm above 25 GeV, and from ~ 40 cm below ten layers to $\lesssim 10$ cm beyond twenty, with the layer count the cleanest single predictor. The error is thus worst exactly where the axis is hardest to fit—sparse, low-energy Tracksters spanning few layers—which is why selecting the single most energetic Trackster per event, the object that actually carries the photon, is what brings the clean 10-to-10 PCA to $\text{RMS}(\Delta R) = 9.91$ cm.

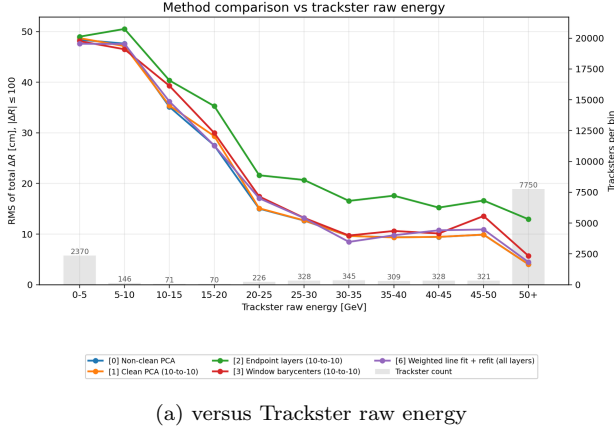
We judged this ~ 10 cm leading-Trackster resolution good enough to bin every reconstruction-side metric that follows, and chose the clean 10-to-10 PCA to compute R . A natural next step, left outside the scope of this project, would be to add a further dimension to the direction fit: each LayerCluster also carries a timing measurement, and a displaced, non-pointing shower

arrives with a time-of-flight pattern that constrains its direction independently of the spatial axis. Exploiting Trackster timing to sharpen R_{reco} is a promising avenue for future work.

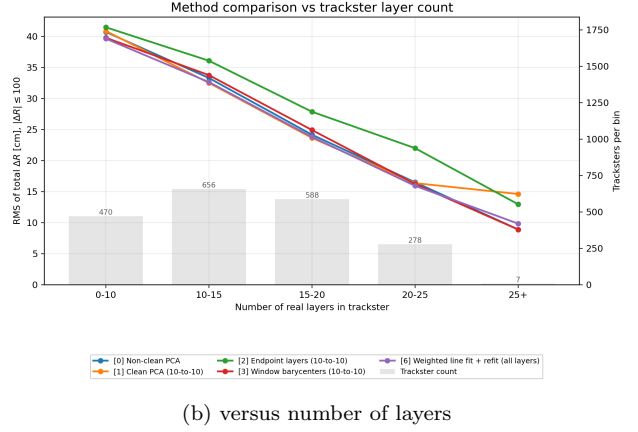
The displacement observables R and α , together with a Trackster-timing observable, were added to the standard HGCAL validation as new per-object quantities so that any efficiency, purity, duplicate or merge plot can be produced as a function of displacement (CMSSW pull request [#51510](#)).

3 Displaced photon gun

To measure the dependence of reconstruction performance on displacement, we need samples in which the shower energy and trajectory can be controlled independently of a particular LLP model. For this purpose, Bruno Alves developed the original displaced-particle gun for the project (CMSSW pull request [#51006](#)). My contribution was to test and improve this generator: I



(a) versus Trackster raw energy



(b) versus number of layers

Figure 5: Displacement-radius resolution $\text{RMS}(\Delta R)$ versus (a) Trackster raw energy and (b) the number of layers the Trackster spans, for each estimator (coloured curves). The resolution improves by a factor of several from low to high energy and from few to many layers, confirming that the error tracks how well the shower direction can be fitted. Grey bars (right axis) give the Trackster count per bin.

revised its sampling and configuration, corrected the production timing, and updated the beamspot choice in the generation workflow. These improvements are collected in pull request [#51489](#).

The studies use photons, whose trajectories are not bent by the magnetic field. The photons are inserted into the simulation immediately before HGCal, so they do not traverse the tracker and convert or shower there. This isolates the response to calorimeter for a non-pointing photon. The gun therefore supplies a controlled test of the reconstruction geometry; it does not simulate the full production and decay of an LLP.

3.1 Trajectory geometry and surface constraints

The construction distinguishes the *sampled origin* from the actual *production vertex* passed to the detector simulation. The origin is a point in the plane $z = 0$,

$$\mathbf{x}_0 = (R \cos \phi_0, R \sin \phi_0, 0), \quad (5)$$

so its radius is exactly the displacement observable defined in Section 2. Given a momentum polar angle θ and azimuth ϕ_p , the straight trajectory is

$$\mathbf{x}_\perp(z) = \mathbf{x}_{0,\perp} + z \tan \theta (\cos \phi_p, \sin \phi_p). \quad (6)$$

Here θ is measured from the positive beam axis, while ϕ_0 specifies the position of the origin and ϕ_p specifies the momentum direction. These two azimuths do not need to coincide. The non-pointing angle α follows from the resulting trajectory and impact point; it is not independently sampled.

The photon is created where this line crosses a configurable *production plane*, placed at $z = 319$ cm in the example configuration. Its position must lie within a specified annulus, optionally restricted in azimuth. An additional *target plane*, near the back of the electromagnetic calorimeter at $z \simeq 362$ cm, can impose a second annular acceptance. Requiring the line to pass through both regions selects trajectories that enter the calorimeter and continue through its volume. Suitable

radial margins can reduce edge leakage, although an accepted trajectory alone does not guarantee containment of the entire shower, whose transverse spread is set by the calorimeter’s Molière radius. The target constraint is optional and assumes straight propagation, so the implementation permits it only for neutral particles.

In particular, neither surface is a second independently sampled vertex. Both constrain the same line from $\mathbf{x}_0$, and the production vertex is obtained by evaluating Eq. (6) at the production-plane position. Consequently, back-extrapolating the generated momentum from that vertex recovers the sampled R , even though the photon is actually introduced into the simulation far downstream of $z = 0$.

3.2 Sampling without biasing the displacement scan

The improved gun supports two distributions for the origin radius. Sampling R^2 uniformly gives a constant density per unit transverse area, with $dN/dR \propto R$. Sampling R uniformly instead gives equal expected populations in equal-width displacement bins. We use the latter for displacement scans. The origin azimuth is sampled uniformly within its configured range.

A simple approach would draw a direction and discard the entire trial whenever it misses either surface. This can be inefficient and, if the origin is also redrawn, changes the accepted displacement distribution according to the geometric acceptance. The improved implementation holds the sampled origin fixed and constructs the allowed polar-angle intervals before drawing θ . For a trial momentum azimuth ϕ_p , write $s = \tan \theta$. At a surface $z = z_k$, the squared transverse radius is

$$r_k^2(s) = R^2 + 2z_k R \cos(\phi_p - \phi_0)s + z_k^2 s^2. \quad (7)$$

The annular requirement $r_{k,\min}^2 \leq r_k^2(s) \leq r_{k,\max}^2$ gives allowed intervals in s . The gun intersects these intervals for the production plane, the optional target plane, and the configured θ bounds. It then samples

uniformly in θ over the surviving intervals, weighting each interval by its angular width.

Momentum azimuths with no allowed interval are retried. Spatial azimuthal cuts on the surfaces are checked after sampling and can also reject a direction. The origin remains fixed throughout: exhausting the configurable attempt limit raises an error instead of silently replacing an inconvenient origin. Geometry checks also reject some incompatible configurations before generation begins. This preserves the requested origin distribution in a successfully generated sample while avoiding many impossible direction trials. It does not imply that the accepted impact positions or the overall θ distribution are uniform; both reflect the surface constraints.

The momentum magnitude is sampled after a valid trajectory has been found. The configuration explicitly selects either a uniform total-energy distribution or a uniform transverse-momentum distribution, together with the corresponding range. For photons these choices are related by

$$E = \frac{p_T}{\sin \theta}, \quad (8)$$

in units with $c = 1$. Thus, a fixed p_T range produces increasingly energetic photons at small θ . Sampling E directly allows the energy distribution to be held independent of the accepted direction, which is useful when isolating displacement effects.

3.3 Timing and validation of the pointing control

Before interpreting displaced-shower failures, we tested the pointing limit $R = 0$ against the existing CloseBy gun. Each event contains a single generated photon, so ideally its shower would be reconstructed as exactly one Trackster; a larger multiplicity signals fragmentation of that shower. At $R = 0$ the displaced gun generates pointing trajectories and should reproduce the CloseBy response when the kinematics and generation conditions are matched. Instead, it initially produced substantially more Tracksters per event than CloseBy, revealing excessive fragmentation even before any displacement was introduced. Investigating this discrepancy exposed problems in the sample setup.

The first was the production time. The original gun assigned $t = 0$ to a photon created immediately before HGCAL, ignoring the flight time required to reach that point. Its signals therefore arrived earlier than a particle travelling at the speed of light from the interaction point could have reached the calorimeter. Such arrival times are incompatible with the timing assumptions of the hit-reconstruction chain, which rejected affected signals. The resulting reconstructed energy was both severely reduced and unevenly retained across the shower. This disrupted the energy deposits available for clustering, breaking a single photon shower into several reconstructed Tracksters and inflating the multiplicity. It was therefore a consequence of the incorrect generation time, rather than evidence of a

displacement-induced clustering failure at $R = 0$. The corrected implementation assigns a time from the path length and particle velocity. More precisely, its present convention is

$$ct_{\text{prod}} = |\mathbf{x}_0| + \frac{|\mathbf{x}_{\text{prod}} - \mathbf{x}_0|}{\beta}, \quad \beta = \frac{|\mathbf{p}|}{E}, \quad (9)$$

where the expression for β uses $c = 1$. The first segment, from the nominal interaction point to the sampled origin, is assumed to be traversed at the speed of light; the second uses the generated particle's velocity. For photons, $\beta = 1$. This is a prescribed timing convention rather than a sampled LLP lifetime. At $R = 0$ it reduces to the ordinary photon flight time to the production plane, about 11 ns for these geometries. Correcting the time partially removed the anomalous behaviour of the pointing control.

A second difference was the sampled momentum variable. The displaced gun used p_T sampling, whereas the CloseBy reference sampled energy. By Eq. (8), the former generated much higher energies at large pseudorapidity, where the showers split into more Tracksters. Introducing direct energy sampling removed most of the remaining multiplicity discrepancy (Fig. 6). Agreement is close over most of the displayed range, with some differences in the highest- η bins; the comparison does not establish exact equivalence of the two samples.

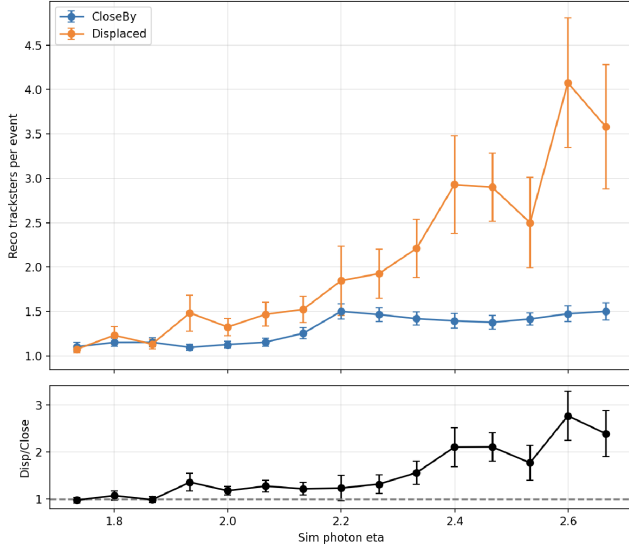
Finally, the displaced-particle workflow used the DBrealisticHLLHC beamspot smearing, which moved the carefully placed production vertices and affected the truth-particle selection near the HGCAL boundary. The workflow was changed to the CloseBy beamspot prescription appropriate to particles generated at the calorimeter. Together, the timing, energy-sampling and beamspot corrections established a usable pointing reference for the displacement scans that follow.

4 Reconstruction performance metrics

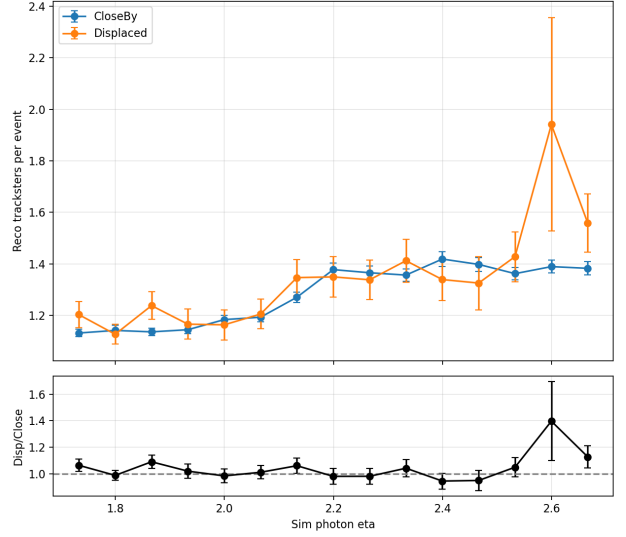
The Trackster multiplicity of Section 3 is a useful single-photon diagnostic, but it says nothing about how the shower energy is distributed among reconstructed objects, nor about contamination when several showers overlap. We therefore use the HGCAL validation framework, which associates reconstructed Tracksters to SimTracksters by shared calorimeter content and from that defines efficiency, duplication, merging and fake rates.

4.1 Shared energy and association scores

The association uses common LayerCluster content, not proximity of the reconstructed and simulated axes. Let e_i^S and e_i^T be the energies of LayerCluster i in a SimTrackster S and a reconstructed Trackster T ; truth assignments may be fractional, whereas in CLUE3D a LayerCluster is never shared between Tracksters, so



(a) Displaced gun sampling p_T



(b) Displaced gun sampling energy

Figure 6: Pointing-control comparison of the CloseBy and displaced guns. The upper panels show the mean number of reconstructed Tracksters per event versus simulated-photon pseudorapidity; each event contains one generated photon, so ideal reconstruction would give a multiplicity of one. Values above one indicate shower fragmentation. The lower panels show the displaced-to-CloseBy ratio. These plots compare the energy-sampling choices, rather than the timing correction.

e_i^T is either zero or the full LayerCluster energy. The shared energy and the recovered fraction are

$$E_{\text{shared}}(S, T) = \sum_i \min(e_i^S, e_i^T), \quad (10)$$

$$f_{ST} = \frac{E_{\text{shared}}(S, T)}{E_S}, \quad E_S = \sum_i e_i^S, \quad (11)$$

so f_{ST} measures how much of the true shower—referred to the SimTrackster energy E_S , not the generated photon energy—is collected in a single reconstructed object. Two directional scores quantify the mismatch in each direction,

$$s_{S \rightarrow T} = \frac{\sum_{i \in S} [\max(0, e_i^S - e_i^T)]^2}{\sum_{i \in S} (e_i^S)^2}, \quad (12)$$

$$s_{T \rightarrow S} = \frac{\sum_{i \in T} [\max(0, e_i^T - e_i^S)]^2}{\sum_{i \in T} (e_i^T)^2}. \quad (13)$$

Lower is better: $s_{S \rightarrow T}$ penalizes missing SimTrackster content, while $s_{T \rightarrow S}$ penalizes reconstructed content foreign to S . Because they weight squared energies, a score cut is not equivalent to a cut on f_{ST} .

4.2 Efficiency, duplicate rate and merge rate

A SimTrackster is counted as reconstructed if some Trackster holds more than half of its energy. In a bin B of a truth observable,

$$\epsilon(B) = \frac{N_S(B; \exists T : f_{ST} > 0.5)}{N_S(B)}. \quad (14)$$

Each SimTrackster counts at most once and its fragments are not summed, so efficiency measures energy

recovery in one object, not that the shower is reconstructed as a single piece. The duplicate rate counts showers reconstructed more than once, and the merge rate—defined from the reconstructed side—counts objects matching more than one truth shower,

$$D(B) = \frac{N_S(B; \#\{T : s_{S \rightarrow T} < 0.6\} \geq 2)}{N_S(B)}, \quad (15)$$

$$M(B) = \frac{N_T(B; \#\{S : s_{T \rightarrow S} < 0.6\} \geq 2)}{N_T(B)}, \quad (16)$$

where $N_T(B)$ counts the nonempty reconstructed Tracksters. All three use the working score threshold 0.6, an explicit override of the validation default 0.2. Small fragments can fail the association cut, so a low duplicate or merge rate does not by itself exclude fragmentation—a caveat revisited in the two-shower study.

4.3 Fake rate, purity and displacement binning

A reconstructed Trackster is non-fake if it has at least one SimTrackster match with $s_{T \rightarrow S} < 0.6$; the fake rate is the complementary fraction, and stays near zero even for fragments fully attributable to one shower. The validation's *purity* is instead a truth-side rate—the fraction of SimTracksters with a match at $s_{S \rightarrow T} < 0.2$ —and should not be confused with the per-object energy fraction E_{shared}/E_T .

Efficiency, duplicate and purity rates use the SimTrackster denominator and are binned in truth R or α ; fake and merge rates use the reconstructed-Trackster denominator and the barycentre-plus-clean-PCA displacement of Section 2, with no truth match required,

so they inherit its resolution. Under pileup a signal-only rate counts only the signal photon's SimTrackster, whereas an inclusive rate also counts pileup SimTracksters; the sample and truth selection must therefore be stated with every curve.

4.4 Improvements to the validation framework

Applying these metrics to isolated displaced showers required changes to the standard validation. A restrictive sim-to-reco threshold can hide splitting, because two fragments need not each resemble the full shower closely enough to pass. I therefore made the non-fake, merge, purity and duplicate score thresholds independently configurable, so the duplicate threshold can be relaxed without touching the energy-based efficiency definition. Together with the displacement and timing observables and a ROOT segmentation-fault fix in separate-plot production, these changes went into CMSSW pull request [#51510](#).

A subsequent refactoring of histogram booking and filling (pull request [#51565](#)) extended the validation to two-dimensional distributions. Recording numerator and denominator jointly in, for example, R and energy or R and pseudorapidity allows performance to be evaluated in slices of the second variable—essential for distinguishing a true displacement dependence from changes in the energy or angular composition of the sample.

5 How CLUE3D works

CLUE3D is the existing CMS algorithm that groups LayerClusters across calorimeter layers into Tracksters using the density-based idea of CLUE [3]. We will summarize the ingredients needed to interpret the performance studies: its projected transverse distance, local density, and seed-follower assignment. The studies use the default HLT CLUE3D configuration without tuning for displaced showers, except where a parameter variation is explicitly stated. The transverse scales quoted below apply to silicon LayerClusters, which contain the electromagnetic shower core.

5.1 Neighbourhoods projected from the interaction point

Let LayerCluster i have position (x_i, y_i, z_i) , transverse radius $r_i = \sqrt{x_i^2 + y_i^2}$, azimuth ϕ_i , layer index ℓ_i , and energy E_i . The algorithm stores

$$q_i = \frac{r_i}{|z_i|}. \quad (17)$$

To compare a candidate j with i , it projects j onto the plane of i along the ray from the origin through j . The projected radius is

$$r_{j \rightarrow i} = |z_i| q_j = \frac{|z_i|}{|z_j|} r_j. \quad (18)$$

The squared transverse distance used by the implementation is then

$$d_{XY}^2(i, j) = (r_i - r_{j \rightarrow i})^2 + r_{j \rightarrow i}^2 (\Delta\phi_{ij})^2, \quad (19)$$

where $\Delta\phi_{ij}$ is the azimuthal difference wrapped into $[-\pi, \pi]$. The second term uses the small-angle arc distance in azimuth. This is a distance evaluated on the reference layer, rather than the ordinary three-dimensional separation of the two LayerClusters; exchanging i and j changes the reference plane and can change its value, so $d_{XY}(i, j) \neq d_{XY}(j, i)$.

Candidate searches are restricted to the same end-cap, within three layers before and after the reference layer. They also use nearby position-based (η, ϕ) tiles: up to two tiles on either side for density estimation, and one for the nearest-higher search. These windows select candidates before the projected distance is evaluated.

5.2 Density and Trackster formation

The local density of LayerCluster i is

$$\rho_i = E_i + 0.2 \sum_{j \in \mathcal{N}_i} E_j, \quad (20)$$

where $\mathcal{N}_i$ contains candidate LayerClusters on other layers with $d_{XY}(i, j) < d_c$, and $d_c = 1.8$ cm. The algorithm then finds the nearest higher-density candidate $h(i)$ on another layer, minimizing the projected transverse distance, and records $\delta_i = d_{XY}(i, h(i))$. This search can find neighbours beyond d_c ; if none is found, the distance is treated as infinite.

LayerClusters are classified as follows:

- **Seeds** start new Tracksters. They require $\rho_i \geq \rho_c = 0.6$ GeV, a self-energy fraction $E_i/\rho_i > 0.15$, and separation from higher-density activity: $\delta_i > d_m = 1.8$ cm or a nearest-higher layer separation exceeding five layers. Within the three-layer search window, the latter condition matters only when no higher neighbour is found.
- **Outliers** have low density, $\rho_i < \rho_c$, and a distant nearest-higher neighbour, $\delta_i > 2d_m = 3.6$ cm.
- **Followers** are the remaining LayerClusters and inherit the Trackster assignment of their nearest-higher neighbour, if one exists.

The density radius d_c and seed distance d_m are separate parameters, although both default to 1.8 cm. The larger outlier threshold means that d_m is not a universal maximum follower-link length.

Starting from each seed, its Trackster index is propagated through the follower links (Fig. 1b). Each assigned LayerCluster belongs wholly to one Trackster; clusters not connected to a seed remain unassigned. Outlier status propagates the same way: a follower whose nearest-higher chain terminates in an outlier rather than a seed is itself treated as an outlier. Candidates with fewer than two LayerClusters are removed.

The barycentre, PCA direction and timing are computed after clustering, so the fitted shower direction does not enter the neighbour search. The next section examines the performance of this default algorithm on displaced photons.

6 CLUE3D performance for displaced photons

The central question of this project is whether CLUE3D can reconstruct a photon shower whose trajectory does not point back to the interaction point. The single-photon study reveals both performance loss and an unexpected structure: the loss develops in two stages, separated by a plateau. The evidence identifies fragmentation induced by the origin-based projection as the mechanism behind the observed loss.

6.1 The performance loss is not a smooth decline

The mechanism study uses single displaced photons without pileup, with Run4D121 geometry and the default HLT CLUE3D configuration. The wide-range production samples R uniformly from 0 to 350 cm and photon energy from 10 to 250 GeV, subject to the production- and target-surface constraints described in Section 3. The initial investigation was used 50 000 events, one event containing one displaced photon. Individual diagnostic plots impose their own requirements, stated where relevant, rather than necessarily containing identical numbers of objects.

In addition to the efficiency of Section 4, I recorded the largest fraction of each SimTrackster’s energy captured by any one reconstructed Trackster,

$$f_{\max}(S) = \max_T \frac{E_{\text{shared}}(S, T)}{E_S}, \quad (21)$$

with zero assigned when there is no associated reconstructed object. Efficiency is the fraction of SimTracksters with $f_{\max} > 0.5$; the mean $\langle f_{\max} \rangle$ measures energy recovery continuously, without reducing each shower to a pass or fail.

Figure 7 shows that the standard efficiency is close to unity at small displacement, falls to roughly 85% in the intermediate region, and reaches about 20% near the end of the scan. The mean captured fraction makes the structure particularly clear: it stays near 0.92 up to approximately $R = 180$ cm, falls to about 0.78 over 180–220 cm, remains near that value to about 270 cm, and then falls steeply to roughly 0.38 near 350 cm.

The intermediate plateau was the main puzzle. Increasing the displacement should progressively violate the pointing assumption, yet the reconstruction temporarily stops deteriorating. Varying the shared-energy requirement changes the numerical efficiency and the visibility of the plateau. However, the two-stage loss remains in the continuous mean $f_{\max}$, so it cannot be explained by the choice of the 0.5 efficiency threshold.

6.2 Testing geometric explanations

I first investigated whether the limited (η, ϕ) tile search could lose LayerClusters from the same shower, since a displaced shower does not keep a constant (η, ϕ) from layer to layer and its LayerClusters could drift out of the search window. The event displays and distributions of layer-to-layer angular steps did not support this as the main explanation. Across the full displacement scan, even the largest single-layer step of a shower stays well below the tile reach (Fig. 9), so the neighbour always falls inside the tile search. The distinction matters: the candidate can be found by the tile search and still be too distant under the projected metric.

Figure 8 illustrates this directly. At $R \simeq 297$ cm, one SimTrackster is split into five reconstructed Tracksters, and the largest shared-energy fraction is only 0.21. The successive colours follow different longitudinal pieces of a continuous shower core. Several changes of colour occur while the core remains within one ϕ tile. This is fragmentation of one shower, not five independently produced photons, and—consistent with the sample-wide picture of Fig. 9—no tile boundary is crossed.

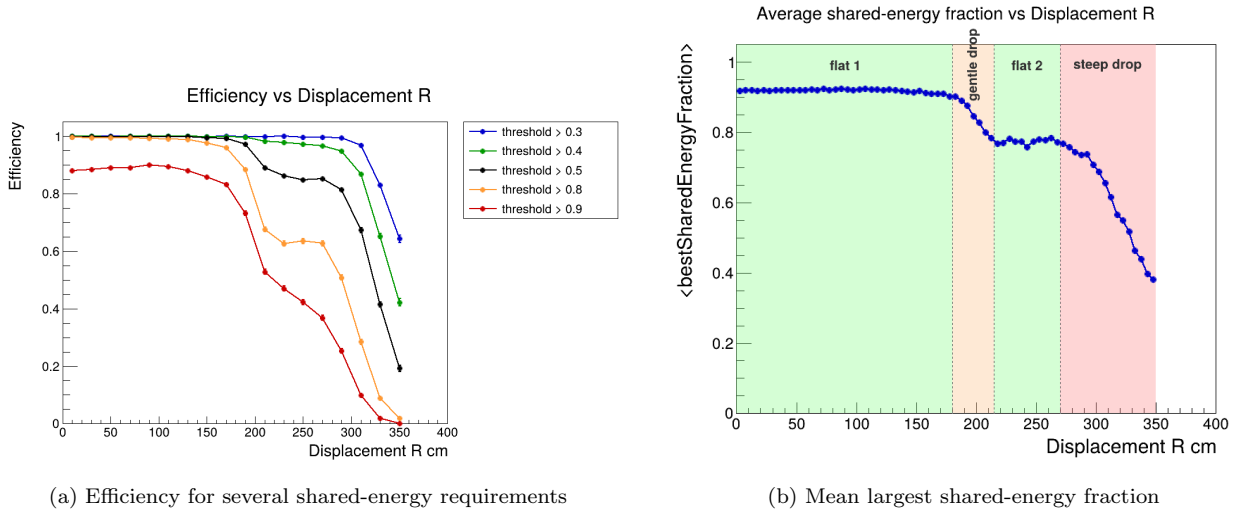


Figure 7: Single-photon reconstruction without pileup. (a) Efficiency for several shared-energy requirements; the black curve is the standard $f_{\max} > 0.5$. (b) Mean $f_{\max}$, showing the two-stage loss and the intermediate plateau. Shaded bands are approximate slope regions.

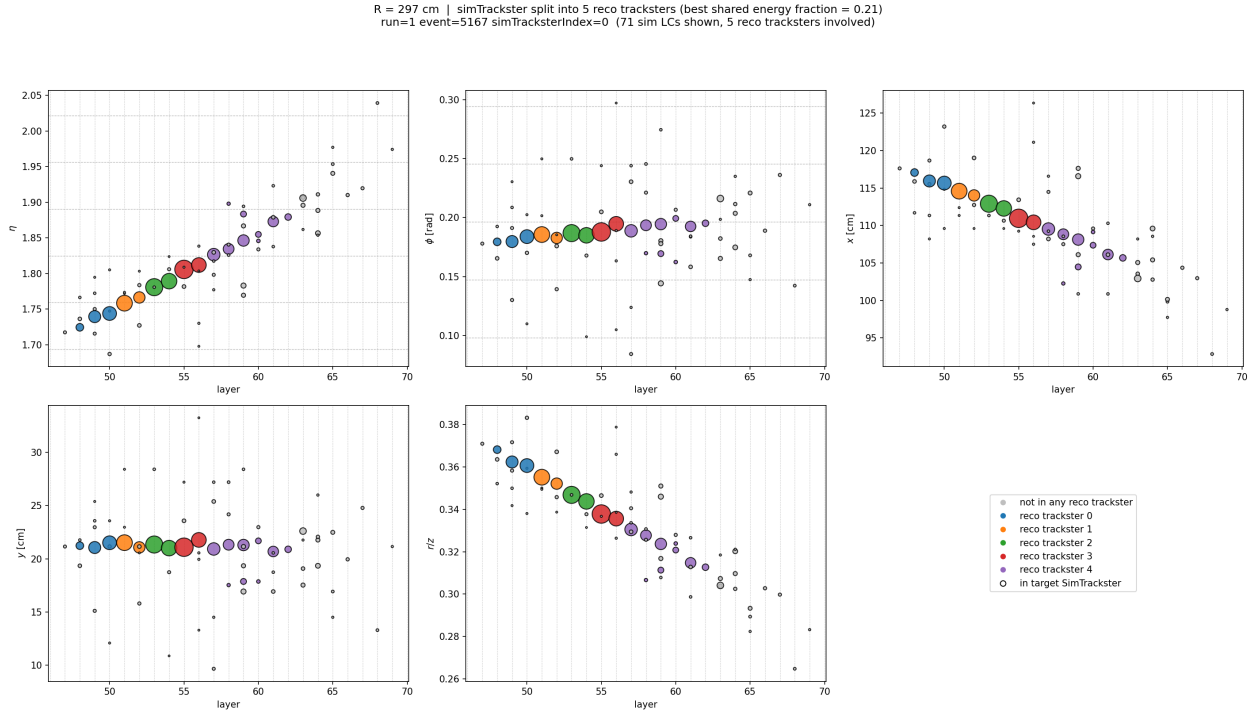
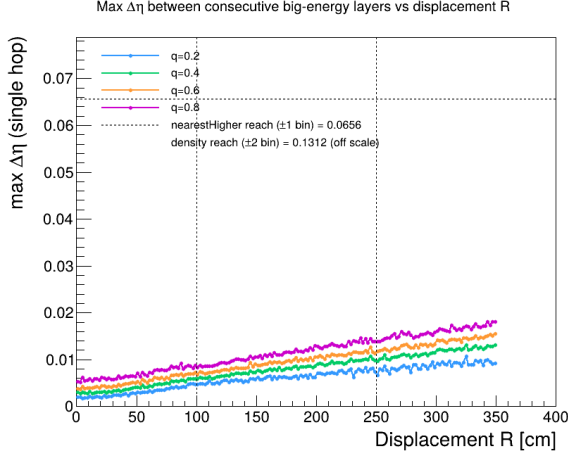
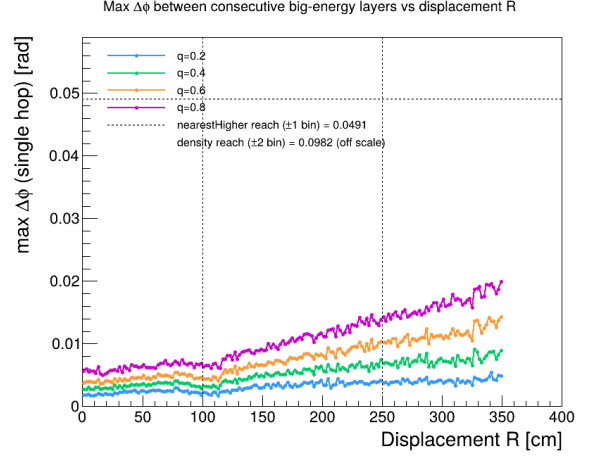


Figure 8: A single photon at $R \approx 297$ cm split into five reconstructed Tracksters. LayerClusters versus layer index in position-based η , ϕ , x , y and r/z ; colour identifies the reconstructed Trackster and marker size the energy. Outlined markers belong to the target SimTrackster, grey markers are unassigned, and the angular panels include tile boundaries. The shower core stays continuous in physical space while its assignment changes repeatedly.

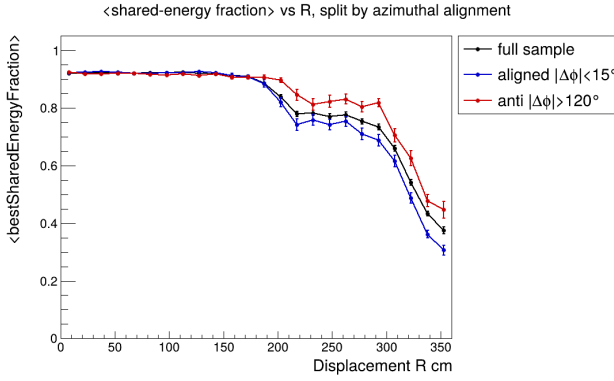


(a) Largest single-layer step in η .

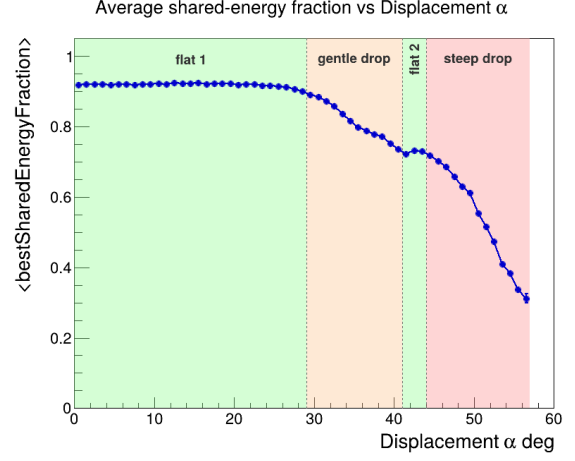


(b) Largest single-layer step in ϕ .

Figure 9: Sample-wide test of the tiling explanation. For each shower, the largest angular step between consecutive high-energy layers—the worst hop the tile search must bridge—versus R (quantiles $q = 0.2$ – 0.8). The dashed line is the nearest-higher tile reach (± 1 tile); the density reach (± 2 tiles) is off scale. Even the worst hop stays well within the tile reach at every displacement, so lost LayerClusters cannot be blamed on the (η, ϕ) tiling and must arise from the projected distance metric.



(a) Separation by azimuthal alignment



(b) Mean captured fraction versus α

Figure 10: Checks of trajectory mixing. (a) Aligned ($|\Delta\phi| < 15^\circ$) and oppositely oriented ($|\Delta\phi| > 120^\circ$) photons, with $\Delta\phi$ between the origin at $z = 0$ and the target-plane position; the aligned subset still shows the intermediate plateau. (b) The same observable versus the non-pointing angle α , with a shorter plateau. The gun’s surface constraints correlate R and α .

A second possibility was that scalar R mixes different trajectory orientations. I separated photons by the azimuthal difference between their back-extrapolated origin and their position at the target plane. The aligned and oppositely oriented populations do have different captured fractions at the same R , but the plateau survives within the aligned population (Fig. 10). Mixing these populations therefore does not create the feature. Binning in α also reveals a short plateau, around 41° – 44° , followed by a strong decline. A further split into broad α ranges retained the flat- R region within the $29^\circ \leq \alpha < 49^\circ$ group (Fig. 11). These checks show that the feature survives changes of projection and sample subdivision, while also establishing that R alone does not specify every aspect of the trajectory.

6.3 The geometric origin of the growing distance

The projected distance of Section 5 suggests a direct mechanism. For a straight trajectory on the positive endcap, write its transverse position as

$$\mathbf{x}_\perp(z) = \mathbf{b} + z\mathbf{a}, \quad R = |\mathbf{b}|, \quad (22)$$

where $\mathbf{a}$ is the transverse slope and $\mathbf{b}$ the intercept at $z = 0$. For a pointing trajectory $\mathbf{b} = 0$: both r/z and ϕ are constant along the ideal shower axis. For a displaced trajectory, $\mathbf{x}_\perp(z)/z = \mathbf{a} + \mathbf{b}/z$ changes with depth, as the r/z panel of Fig. 8 demonstrates for a real shower.

Projecting a point on layer j to layer i along a ray from the interaction point gives

$$\mathbf{x}_{j \rightarrow i}^{\text{proj}} = \frac{z_i}{z_j} \mathbf{x}_\perp(z_j). \quad (23)$$

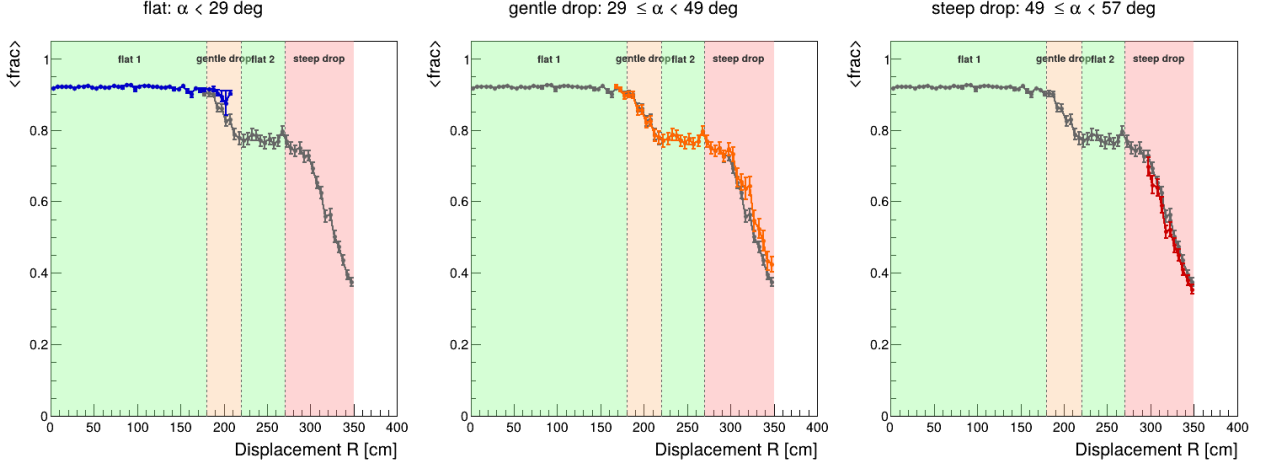


Figure 11: Mean captured fraction versus R , split into three non-pointing bands: flat ($\alpha < 29^\circ$), gentle drop ($29^\circ \leq \alpha < 49^\circ$) and steep drop ($49^\circ \leq \alpha < 57^\circ$). The grey curve is the full-sample average in each panel. The “flat 2” plateau near $R \simeq 250$ cm persists in the 29° – 49° band, so it is not an averaging artefact.

Subtracting this projected position from the true position on layer i cancels the slope term exactly:

$$\Delta \mathbf{x}_{ij} = \mathbf{x}_\perp(z_i) - \mathbf{x}_{j \rightarrow i}^{\text{proj}} \quad (24)$$

$$= \mathbf{b} \left(1 - \frac{z_i}{z_j} \right) = \mathbf{b} \frac{z_j - z_i}{z_j}. \quad (25)$$

Thus the exact Cartesian offset between ideal axis points is $|\Delta \mathbf{x}_{ij}| = R|z_j - z_i|/z_j$. For neighbouring layers, and using the small-angle transverse metric implemented in CLUE3D,

$$d_{XY}(i, j) \simeq R \frac{|\Delta z|}{z}. \quad (26)$$

There is a distinction between the exact vector relation and the approximate d_{XY} relation: the code uses radial and azimuthal components, and actual LayerClusters fluctuate around the ideal axis.

Equation (26) explains why the distance grows with both displacement and layer separation even for a perfectly straight photon. It also gives the approximate scale at which a link exceeds the critical distance,

$$R_{\text{crit}} \simeq \frac{d_m z}{|\Delta z|}. \quad (27)$$

This supplies a prediction to test against the observed regions. It does not by itself predict the full efficiency: whether a particular LC becomes a new seed also depends on its local density, its self-energy fraction and the available higher-density candidates.

6.4 A promising radial-only explanation

The first comparison with the layer-gap distances looked particularly promising. I selected the most energetic LC on each layer, retained it if its energy exceeded 1 GeV, and compared pairs separated by $d = 1, 2, 3$ layers. SimTracksters were required to contain at least four LCs. In the radial component alone, the mean

$d = 3$ offset reaches 1.8 cm near the first performance decline, and the mean $d = 2$ offset reaches it near the final decline (Fig. 12). This suggested an appealing explanation in which successively shorter links were lost as displacement increased.

However, that particular assignment of the two declines does not survive using the actual distance in the algorithm. The azimuthal term also contributes to d_{XY} ; it is not consistently negligible. Once both components are included, the crossing positions change. The radial-only alignment was therefore insufficient evidence: the relevant comparison had to use the full projected distance and identify which higher-density neighbours were associated with actual new seeds.

The local displays illustrate why the full distance matters. In Fig. 12(b), the marked same-shower LC on the preceding layer lies 1.82 cm from the new fragment’s core in projected coordinates, just outside the critical circle. This shows how neighbouring pieces of a continuous shower can fall on opposite sides of the distance criterion.

6.5 The two-layer transition explains the first loss

The decisive connection is shown in Fig. 13. Near $R \simeq 200$ cm, the mean full projected distance for $d = 2$ pairs crosses 1.8 cm. At the same time, the energy carried by reco Tracksters which failed to follow nearestHigher $d = 2$ layers away rises sharply. This connects the geometric threshold to the objects that actually remove energy from the best reconstructed match.

Panel (b) isolates those fragments: for each reconstructed Trackster whose seed lies beyond the 1.8 cm threshold from its higher-density neighbour, I sum the full Trackster energy—the seed and all its followers—and normalize it to the shower energy in each R bin, separated by the layer gap d to that neighbour.

The two-layer contribution rises from almost zero to roughly 15% of the reference shower energy through the first decline. It then levels off over approximately the

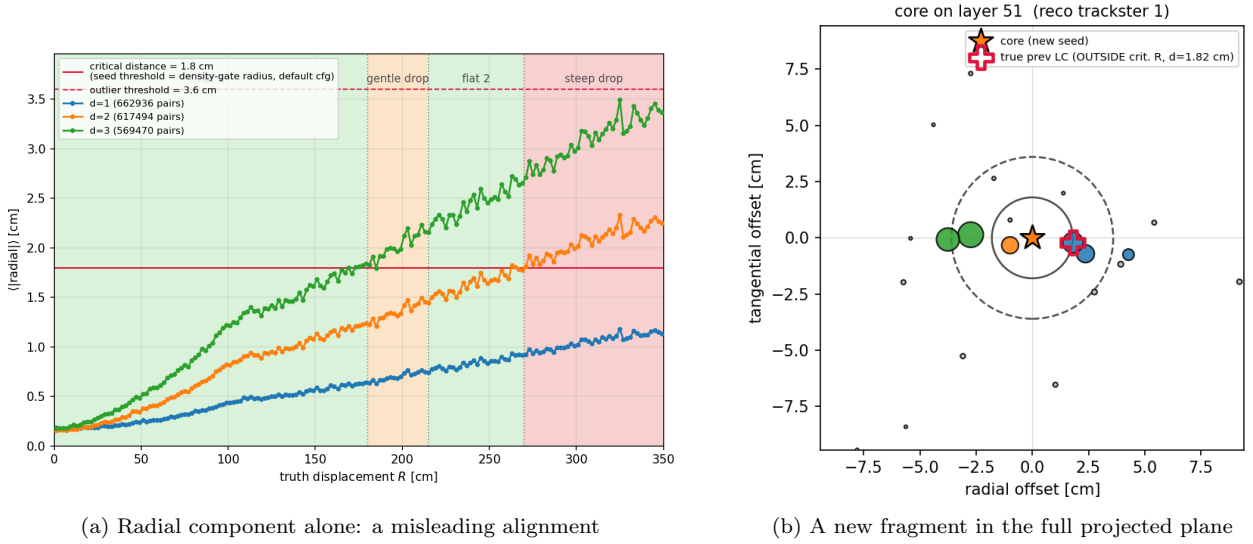


Figure 12: Distinguishing a suggestive correlation from the clustering decision. (a) Radial offsets for representative shower-core LCs separated by d layers; the apparent alignment of the $d = 3$ and $d = 2$ crossings with the two declines omits the azimuthal contribution to the algorithm's distance. (b) A neighbourhood in the full projected coordinates for the $R \simeq 297$ cm event: the star marks the layer-51 core of a new fragment, the cross a same-shower LC on the preceding layer, and the circles the 1.8 and 3.6 cm scales.

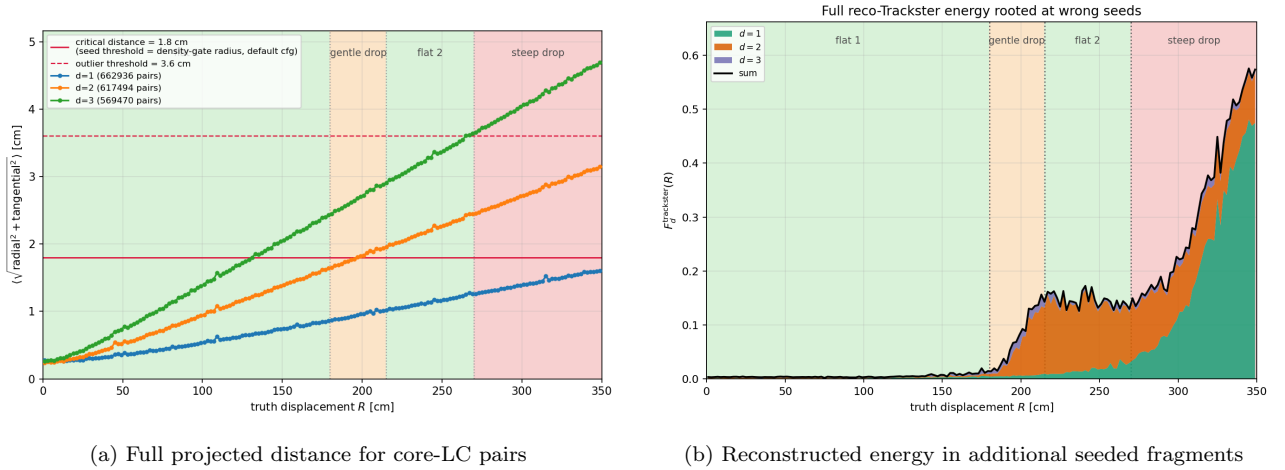


Figure 13: The connection between the geometric threshold and fragmentation. (a) Mean full projected distance for the same core-LC pairs as in the radial-only comparison. (b) Full reconstructed energy in Tracksters whose target-owned seed has a higher-density candidate beyond 1.8 cm, normalized to the SimTrackster energy and separated by the layer gap d to that candidate. The $d = 2$ crossing in (a) coincides with the sharp rise of the $d = 2$ contribution in (b). The geometric pairs in (a) need not be the chosen nearest-higher links.

same displacement range as the plateau in $\langle f_{\max} \rangle$. The plateau therefore corresponds to a region where this newly important source of fragmentation has largely stabilized, before the next one becomes dominant. At larger displacement, the green $d = 1$ contribution rises rapidly and drives the second, stronger loss of captured energy.

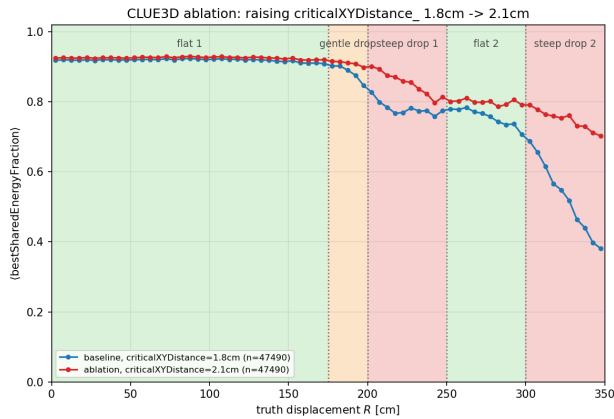
This also explains why the distance plot alone was not enough. Three-layer pairs cross 1.8 cm earliest, but add little seeded energy, because a closer higher-density neighbour can still catch the LayerCluster and prevent a split. And although the mean one-layer distance stays below 1.8 cm for the whole scan, individual one-layer links still exceed it and produce the observed contribution. The distance plot therefore only sets the geometric scale; it is the seed-resolved energy

that shows which crossings actually cause fragmentation, and together they explain the first decline, the plateau and the second decline.

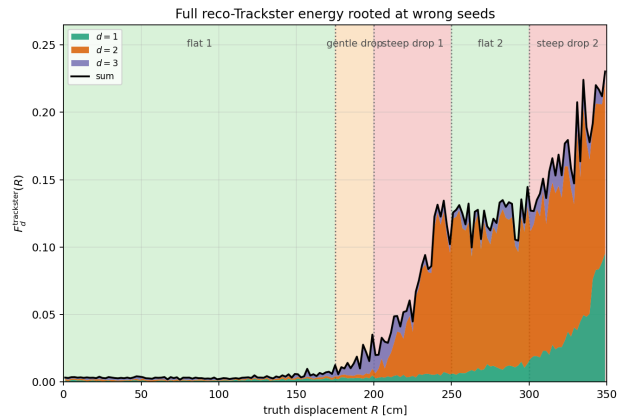
6.6 A controlled change of the critical distance

To test whether this decision actually causes the loss, I reran the HLT reconstruction on the same sample with the critical distance raised from $d_m = 1.8$ cm to 2.1 cm, leaving the density radius at $d_c = 1.8$ cm (the outlier distance, twice d_m , rises automatically to 4.2 cm).

The point is not that a larger d_m gives higher efficiency, but that the same explanation should still predict *where* the degradation happens. For $d_m = 2.1$ cm, Eq. (27) places the $d = 2$ threshold crossing at a larger



(a) Captured energy: default and modified reconstruction



(b) Seeded-fragment energy with $d_m = 2.1$ cm

Figure 14: The controlled 2.1 cm experiment. (a) Captured energy for the default and modified reconstruction, from the same generated sample. (b) Seeded-fragment energy, defined as in Fig. 13(b), showing a later $d = 2$ rise and a strongly reduced $d = 1$ contribution—the shift to larger R predicted by Eq. (27). Its vertical scale differs from the default. Shading describes the modified regimes rather than predicting boundaries.

displacement, so that stage of fragmentation should set in later—and it does: the two-layer seeded contribution turns on at larger R , with the captured fraction and efficiency changing accordingly (Fig. 14(a)).

This explains the mechanism. The origin-based projection pushes apart LCs of the same displaced shower; as displacement grows, their projected distance crosses d_m in stages, creating extra seeds whose followers become separate Tracksters and drain energy from the best match. Changing d_m moves that fragmentation to exactly the displacement the same equation predicts—so the loss is caused by the parameter the geometry implicated, not merely correlated with displacement.

This is a mechanism test, not a tuning of CLUE3D: a larger distance can also merge activity from different showers, so it must be checked with nearby photons and pileup. Still, the single-photon study pins the degradation to a concrete mismatch between the shower geometry and the algorithm’s origin-based distance.

7 Merging in two-photon events

The single-photon study isolates fragmentation, but a useful reconstruction must also keep neighbouring showers separate. Increasing the nearestHigher distance can recover a displaced shower while allowing links to another particle. I therefore extended the displaced-gun studies to two photons and examined both the reconstructed objects and the nearest-higher-density links that form them. All results in this section use the default CLUE3D configuration.

7.1 Controlled shower separation

Two complementary generators control the separation of the photons. In the distance scan, the photons have equal energies and parallel momenta, while their production points are shifted symmetrically by $\pm s/2$ in a randomly chosen transverse direction. The sampled

separations are $s = 1, 2, 4, 8, 16$ cm. Both trajectories must satisfy the gun’s geometrical acceptance. This gives a controlled comparison as the displacement is varied, without changing the opening angle at the same time. The distance study uses 50 000 generated events.

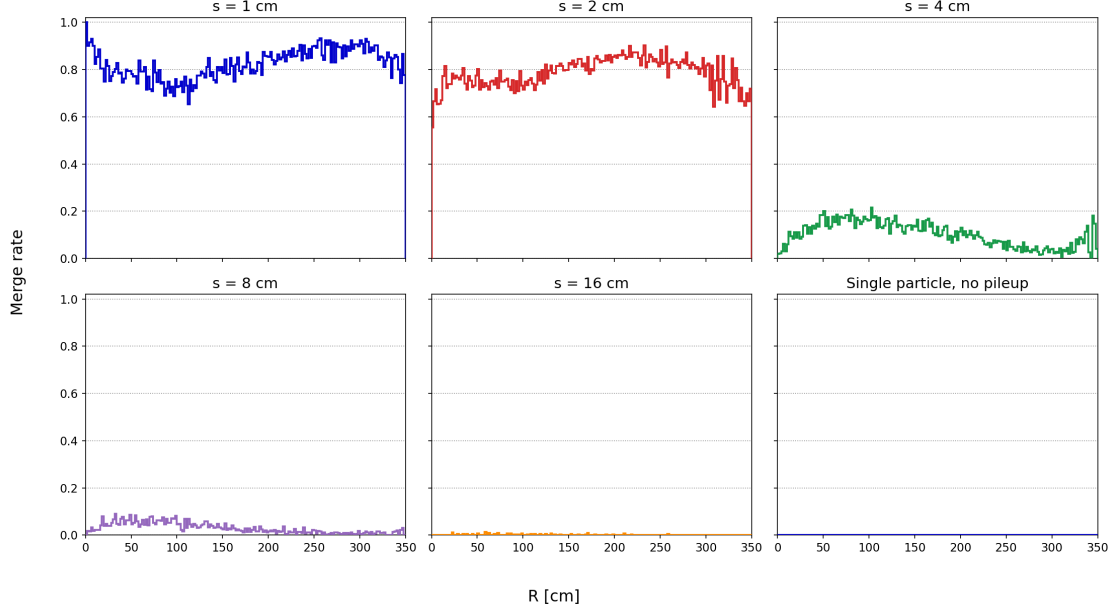
In the opening-angle scan, the two photons instead share a production point and their momenta are rotated symmetrically around a central direction. The angles are $\Delta\theta = 0.001, 0.004, 0.016, 0.064, 0.256$ rad. Their physical separation then grows along the shower depth. The two scans consequently test different geometries: parallel, transversely separated showers and showers that diverge from a common point.

For $s = 4$ cm, the reconstructed-object merge rate initially appears encouraging at large R : after rising at intermediate displacement, it falls again (Fig. 15a). However, this is the same region in which single showers fragment strongly. Splitting increases the number of reconstructed objects in the denominator and can break an object that matched two showers into smaller pieces that no longer pass both association cuts. A decreasing merge rate therefore does not establish that the algorithm has become better at separating the photons.

7.2 A diagnostic based on the clustering links

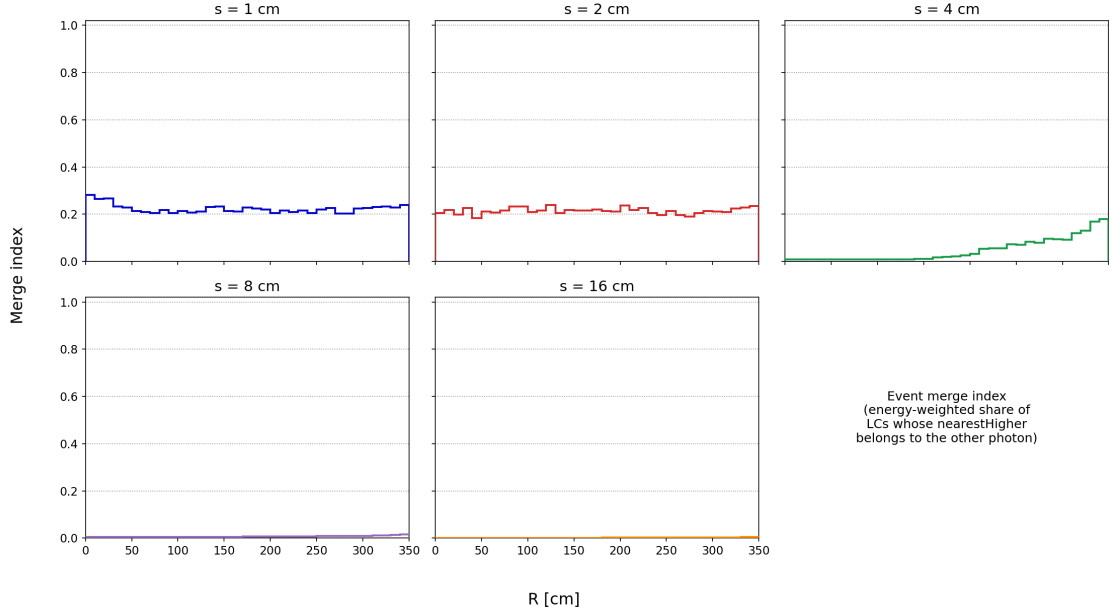
The object-based merge rate is distorted by fragmentation, so I introduced a *merge index* that measures cross-shower mixing directly from the clustering links, without depending on how many Tracksters are produced. Every follower LC i records a nearest-higher neighbour $h(i)$; if $h(i)$ is owned by a different truth shower than i , the algorithm has linked across the shower boundary. The merge index is the energy-weighted fraction of follower energy in such crossing

Two-photon distance scan: TICL Tracksters CLUE3D High — Merge rate



(a) Reconstructed-object merge rate.

Two-photon distance scan: TICL Tracksters CLUE3D High — Event merge index



(b) Energy-weighted cross-shower link index.

Figure 15: Two-photon distance scan with default CLUE3D. Each panel separates the five production-point distances. The object-based rate uses reconstructed R , whereas the link index is binned using the truth displacement of the shower owning the follower LC. In particular, the $s = 4$ cm index increases at large displacement even as the object-based merge rate falls.

links,

$$I_{\text{merge}}(B) = \frac{\sum_{i \in \mathcal{F}_B} E_i \mathbf{1}[o(h(i)) \neq o(i), o(h(i)) \text{ valid}]}{\sum_{i \in \mathcal{F}_B} E_i}, \quad (28)$$

where $o(\cdot)$ is the dominant truth-shower owner (the shower contributing the largest truth energy fraction), E_i the full LC energy, and $\mathcal{F}_B$ the truth-owned followers whose shower passes the truth selection in displacement bin B (seeds and outliers excluded). Energies are pooled across the bin before the ratio is taken, so an index of 0.2 means that 20% of the selected follower energy links into the wrong shower.

The index counts immediate links rather than the total energy ultimately assigned to the wrong shower: one crossing link can redirect a whole chain of followers, and seed and outlier decisions change which LCs enter the denominator. These differences make it complementary to the object-based merge rate.

Figure 15b resolves the apparent improvement. For $s = 4$ cm, the index rises from close to zero near the origin to about 0.18 at large displacement. Cross-shower linking becomes more frequent even while the reconstructed-object merge rate decreases. At $s = 1$ and 2 cm, substantial mixing is already present, with an index of order 0.2; the 8 and 16 cm samples remain much cleaner. An index around 0.2 should not be interpreted as partial success at separating two almost overlapping showers: many links can stay within their own dense core even when both cores ultimately feed the same reconstructed object.

7.3 Connection to the projective geometry

Figure 16 shows an individual cross-shower link, in an event with $s = 4$ cm and $R \simeq 131$ cm. The highlighted red LC belongs to shower B , but its recorded nearest-higher neighbour is the blue LC from shower A , three layers away. The first two panels show why this link is geometrically possible: the two LCs have similar $r/|z|$ and ϕ despite belonging to different showers, so that in the third panel the blue neighbour lies inside the 1.8 cm circle once projected onto the red LC's layer. The algorithm therefore sees a reachable higher-density neighbour across the truth boundary, tying the cross-shower links to the observed merged objects.

The origin projection discussed in Section 6 can also explain how links cross between the showers. Consider ideal parallel axes $\mathbf{x}_A(z) = \mathbf{b}_A + z\mathbf{a}$ and $\mathbf{x}_B(z) = \mathbf{b}_A + \mathbf{s} + z\mathbf{a}$, where $\mathbf{b}_A$ is shower A 's transverse intercept at $z = 0$ and $\mathbf{s}$ is their transverse separation, with $|\mathbf{s}| = s$. Projecting a point on shower B at z_j onto the layer of a point on shower A at z_i gives distance:

$$\mathbf{x}_A(z_i) - \frac{z_i}{z_j} \mathbf{x}_B(z_j) = \mathbf{b}_A \frac{z_j - z_i}{z_j} - \frac{z_i}{z_j} \mathbf{s}. \quad (29)$$

The first term is the displacement-induced offset derived for one shower; the second represents the true separation of the pair. For suitable layer ordering and separation direction, these terms partly cancel. Points

from different showers can then appear close in the projected plane even while points along one shower are pushed apart.

7.4 Which layer gaps produce the merging?

To see whether the three-layer link of Fig. 16 is representative, I split the merge-index numerator by the layer gap $\Delta\ell = |\ell_i - \ell_{h(i)}|$ between each follower and its cross-shower neighbour, keeping the same denominator so the $\Delta\ell = 1, 2, 3$ bands sum to the merge index (Fig. 17, for $s = 2, 3, 4$ cm).

The dominant gap shifts with separation. At $s = 2$ cm, adjacent-layer links carry most of the cross-shower energy and grow with R . At $s = 3$ cm the onset is led by three-layer links, overtaken by two-layer links at larger R . At $s = 4$ cm three-layer links dominate the onset and most of the growth, with two-layer links rising only near the end of the scan. The event display of Fig. 16 therefore shows the link type that drives the onset for the 4 cm pair, not an isolated coincidence.

This ordering follows from how the projection offset scales. Bridging the separation s requires offset, and since the offset grows with the layer gap Δz , a wider gap reaches that offset at smaller R . Wide-gap links therefore start crossing first; as R increases, progressively shorter-gap links reach the same offset and begin merging too. This is the shift towards smaller gaps with displacement seen in the stacked bands, and it explains not just that the merge index rises but which links carry the rise.

7.5 Opening-angle scan

The opening-angle scan gives a complementary result (Fig. 18). For the four smaller angles, the link index is already substantial and remains near 0.2 over most of the displacement range. At the largest angle, 0.256 rad, it starts near zero and increases at large R . A geometrically better-separated pair can therefore also lose separation as displacement grows. Together, these scans show why tuning must consider both shower completeness and cross-shower links: recovering energy in larger Tracksters alone is not sufficient.

8 CLUE3D performance with pileup

Pileup introduces many nearby showers with a broad range of energies and particle species. I studied samples with an average of 200 additional interactions per bunch crossing (PU200), keeping two efficiencies apart: the efficiency of reconstructing the injected signal photon, and the inclusive efficiency measured over all selected in-time SimTracksters (signal plus pileup). This distinction is essential: the signal-only curve tests the same displaced-photon topology in a busier environment, whereas the inclusive curve also changes the population being measured.

$s = 4 \text{ cm} \mid R = 131 \text{ cm} \mid \text{layers } 52\text{-}55 \text{ (3 layers apart)}$

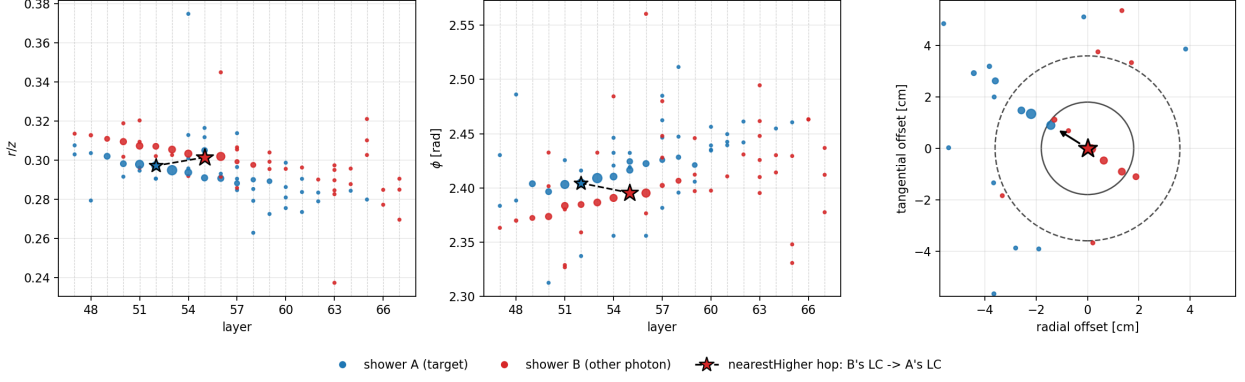


Figure 16: An actual cross-shower nearest-higher link for photons separated by $s = 4 \text{ cm}$ at $R \simeq 131 \text{ cm}$. Blue and red identify the two truth showers. The highlighted LCs occupy layers 52 and 55, but have similar projective coordinates. The right panel shows neighbouring LCs projected onto the red LC's layer; the arrow follows its recorded link to shower A. The solid and dashed circles mark the 1.8 cm critical transverse distance and 3.6 cm outlier distance.

Two-photon distance scan: merge index by cross-shower hop layer gap

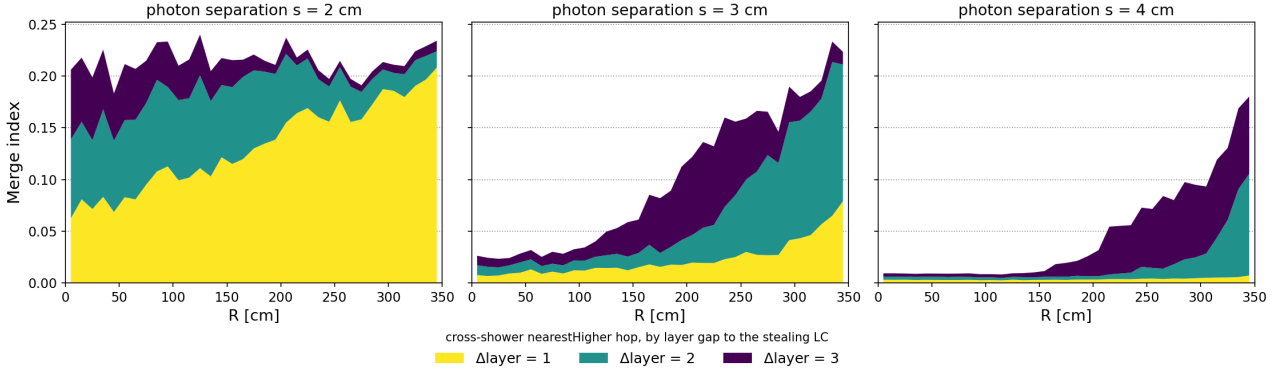


Figure 17: Merge index decomposed by the layer gap of cross-shower nearest-higher links for photon separations $s = 2, 3, 4 \text{ cm}$. Yellow, teal and purple correspond to one-, two- and three-layer gaps. All bands use the same follower-energy denominator, so their heights sum to the merge index. Larger separations favour larger gaps at the onset of mixing; increasing displacement makes smaller-gap contributions more important. The 3 cm panel comes from the finer separation scan.

8.1 Signal efficiency and the meaning of displacement

Figure 19 compares the no-pileup photon sample, the signal photon in PU200, and the inclusive PU200 selection. The first two curves are similar: both retain high efficiency at small displacement and show the large- R degradation identified in Section 6. The inclusive curve is quite different, and it is dominated by pileup: the signal photon accounts for fewer than 1 in 1000 of the selected in-time SimTracksters, so the inclusive denominator is essentially the pileup population rather than the injected photon. It falls sharply just outside the first displacement bin, reaches a minimum, and then increases over a broad range of R . This shape cannot be read as the response to a single photon displaced progressively farther from the beamline. A comparison of the Trackster and TICL-candidate efficiencies also showed no substantial change in this behaviour at the candidate stage.

For pileup particles, resolved R is particularly easy to misinterpret. The boundary position and direction de-

fine a line that can be extrapolated to $z = 0$. Charged particles bend in the magnetic field, so this line is a tangent to a curved trajectory. Its intercept can be far from the beamline even when the particle was produced close to it. A large resolved R in pileup is therefore not evidence for a physically displaced production vertex. The reconstruction consequences, however, are the same either way: a trajectory with a large intercept still projects poorly onto the interaction point, so by Eq. (26) its LayerClusters are pushed apart as $d_{XY} \simeq R|\Delta z|/z$ and suffer the same fragmentation as a genuinely displaced shower.

Even for a straight photon trajectory, the longitudinal spread of collision vertices matters. For a photon produced on the beamline at $z = z_v$, the transverse intercept of its line at $z = 0$ has magnitude

$$R = |z_v| \tan \theta, \quad (30)$$

where θ is the momentum polar angle relative to the beam axis, taking the corresponding acute angle in either endcap. It is not the same experiment as varying the transverse production offset of the displaced gun.

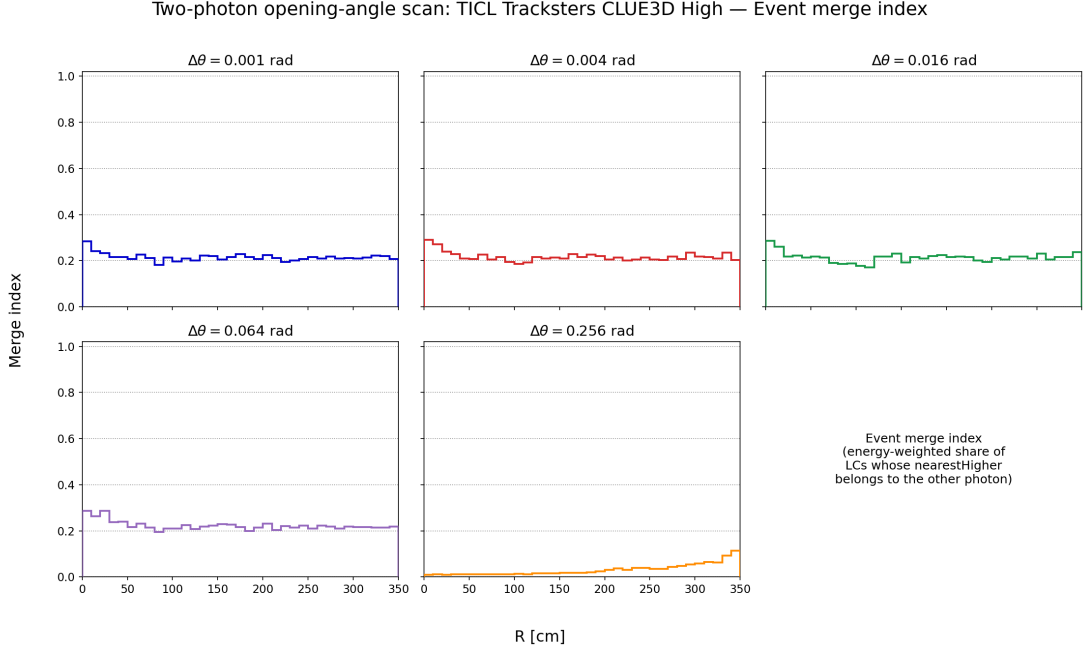


Figure 18: Merge index for the two-photon opening-angle scan. The index uses Eq. (28), pooling follower energies within each bin. The largest opening angle provides the clearest increase in cross-shower linking with displacement.

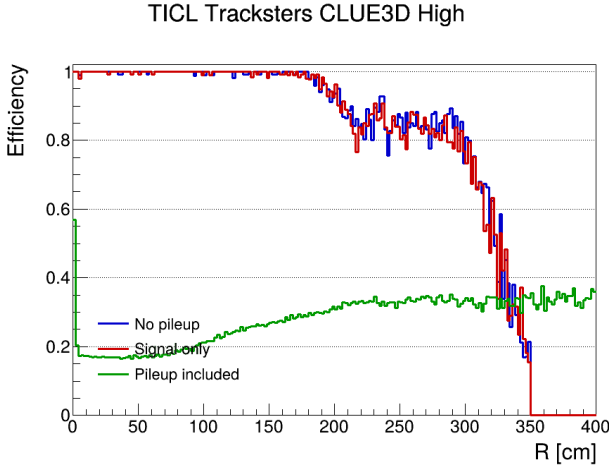


Figure 19: Default CLUE3D efficiency without pileup, for the generated signal photon in PU200, and for the inclusive in-time truth selection in PU200. The inclusive curve has a different denominator and a different interpretation of resolved R for charged particles.

8.2 Separating particle composition from reconstruction effects

To investigate the inclusive shape, I broke the efficiency down per particle in a 20 000-event study, so that numerator and denominator could be examined in the same kinematic slices.

The truth composition changes rapidly near $R = 0$. Photons contribute strongly in the first bins, while charged pions dominate most of the larger resolved-displacement range (Fig. 20). This accounts for part of the inclusive change, but it is not a complete explanation: efficiencies within individual species still depend on R . Photon efficiency falls, whereas the charged-hadron curves tend to rise. The remaining investigation therefore treats these populations separately.

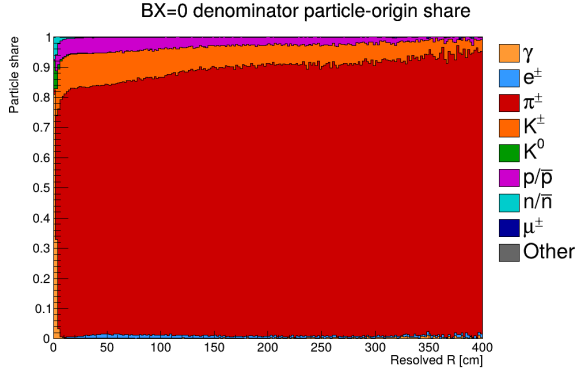
For photons, the loss near the origin is concentrated in the low-energy population, below about 2 GeV. The remaining dependence is really angular, not on displacement. Pileup photons come from vertices close to the beamline, so z_v is small and Eq. (30) makes resolved $R = |z_v| \tan \theta$ essentially a proxy for the polar angle: a small increase in R means a larger θ and hence a rapidly falling $|\eta|$. The two-dimensional efficiency in Fig. 21 confirms this—the structure runs in $|\eta|$ rather than in R , with the loss confined to the lower- $|\eta|$ edge of the HGCal acceptance, where reconstruction is expected to degrade. A one-dimensional photon-versus- R curve therefore inherits this acceptance edge and only looks like a displacement effect.

8.3 Charged showers and LayerCluster multiplicity

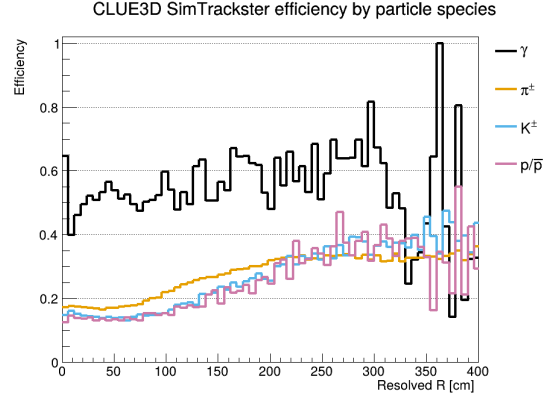
For charged pions, kaons and protons the efficiency instead *rises* with resolved R , the opposite of the photon trend, and the controlling variable is the number of LCs in the truth shower: showers with many LCs rarely have more than half of their energy collected in a single Trackster (Fig. 22). As R grows, the whole pion LC-multiplicity distribution shifts downward—not just its tail—so the population moves towards smaller showers, and the charged-particle efficiency rises with it.

Higher-momentum hadrons bend less, so they keep a smaller resolved intercept R while also developing larger, higher-multiplicity showers; larger R therefore preferentially selects the lower-energy, few-LC showers that CLUE3D reconstructs more easily. This is a selection effect on the pileup population, not a direct action of transverse displacement.

As an aside, similar connection first appeared the other way around. In two-dimensional (R , energy) and (R , $|\eta|$) efficiency maps for pions, bins containing more



(a) Composition of the in-time truth denominator.



(b) Efficiency split by particle species.

Figure 20: Particle composition and reconstruction efficiency versus resolved R in pileup. The changing mixture contributes to the inclusive curve, but does not explain the displacement dependence within each species.

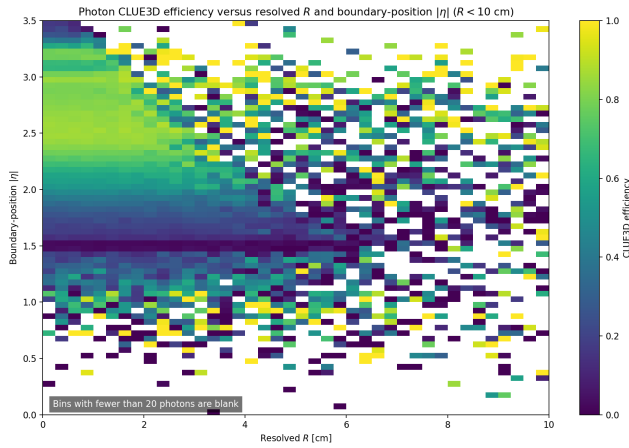


Figure 21: Pileup-photon efficiency versus resolved R and boundary-position $|\eta|$ near the origin. Bins containing fewer than 20 photons are blank. The prominent angular structure motivates separating acceptance and kinematic effects from a dependence on displacement.

reconstructed objects tended to have lower efficiency. The natural guess was local crowding: that such showers sat in busier neighbourhoods, where a high density of nearby reconstructed objects might merge activity or otherwise disrupt the clustering. When I measured this directly, however, efficiency showed no dependence on the number of reconstructed objects in a shower's own neighbourhood, even at fixed R , energy and $|\eta|$.

Anyways, inclusive pileup efficiency must be read together with particle identity, energy, angle and shower size, and the signal-only selection remains the direct test of displaced-photon reconstruction in pileup.

9 Comparison with CLUEstering

CLUEstering is a generalized, heterogeneous implementation of the CLUE approach, with configurable coordinates and distance metrics. It retains the density, nearest-higher and seed-follower structure introduced in Section 5, while allowing the neighbourhood geometry to be changed. My work was to run

and diagnose its application to displaced showers in CMSSW, comparing several configurations with default CLUE3D. The framework can use Cartesian positions directly or projective coordinates; removing the origin assumption is therefore one possibility, rather than a defining property of CLUEstering.

9.1 Cartesian distance metrics

For Cartesian coordinates (x, y, z) , I explored three familiar distance families. Writing $\Delta x = x_i - x_j$, and similarly for y and z , their unweighted forms are

$$D_2 = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}, \quad (31)$$

$$D_1 = |\Delta x| + |\Delta y| + |\Delta z|, \quad (32)$$

$$D_\infty = \max(|\Delta x|, |\Delta y|, |\Delta z|). \quad (33)$$

Euclidean distance D_2 measures straight-line separation, so a fixed distance cut selects a sphere. Manhattan distance D_1 adds the separations along the three axes and selects an octahedron. Chebyshev distance D_∞ takes the largest component and selects a cube: all three coordinate differences must individually lie within the cut. These shapes give different neighbours even when the numerical distance setting is identical.

Weights allow the longitudinal and transverse reaches to be adjusted. With positive weights w_k and coordinate differences Δx_k , the corresponding forms are

$$D_{2,w} = \sqrt{\sum_k w_k (\Delta x_k)^2}, \quad (34)$$

$$D_{1,w} = \sum_k w_k |\Delta x_k|, \quad (35)$$

$$D_{\infty,w} = \max_k (w_k |\Delta x_k|). \quad (36)$$

The weighted examples shown below use $(w_x, w_y, w_z) = (1, 1, 2)$ for Euclidean and Chebyshev distance; the Manhattan example is unweighted. Increasing the longitudinal weight shortens the allowed reach in z . In these conventions, a weight of two reduces the pure longitudinal reach by $\sqrt{2}$

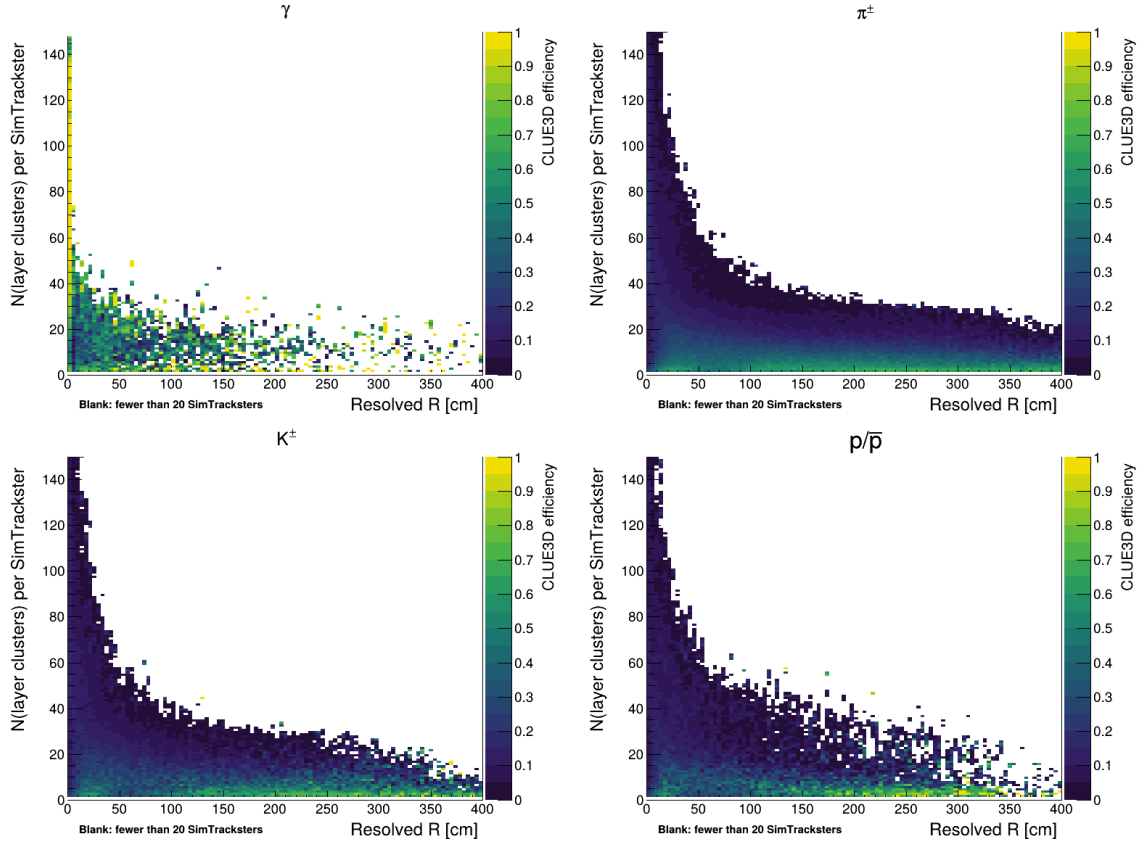


Figure 22: Efficiency in bins of resolved R and truth-shower LC multiplicity, separated by particle species. The charged-hadron panels show lower efficiency for larger LC multiplicities and fewer large showers at large resolved R . Blank bins contain fewer than 20 SimTrackers.

for Euclidean distance and by two for Chebyshev distance. All these Cartesian metrics depend only on coordinate differences and make no projection towards the interaction point.

9.2 A projective metric under development

I also tested a work-in-progress projective configuration, labelled *Gnomonic* in the plots. It maps LC positions to the direction coordinates

$$\mathbf{g} = \left(\frac{x}{|z|}, \frac{y}{|z|} \right), \quad (37)$$

and uses the layer index ℓ as the longitudinal coordinate. Within one endcap, the tested cylinder distance is

$$D_{\text{cyl}}(i, j) = \max \left(\frac{|\mathbf{g}_i - \mathbf{g}_j|}{\sigma_T}, \frac{|\ell_i - \ell_j|}{L} \right). \quad (38)$$

A cut $D_{\text{cyl}} < d$ selects a transverse disc of radius $d\sigma_T$ and a longitudinal interval of half-height dL : a cylinder in the transformed coordinate space. The transverse and longitudinal scales can thus be set separately. Points on an ideal pointing shower have constant $\mathbf{g}$; non-pointing showers drift across this plane, so some degradation with displacement is still expected.

The tested implementation clusters only the electromagnetic section (EE). It uses $\sigma_T = 0.003$, $L = 4$ layers, density radius $d_c = 1$, seed distance 2.8 and outlier

distance 3.6, with the distances expressed in the normalized units of Eq. (38), not centimetres. To account for PU η -uniform density, density threshold also varies with direction:

$$\rho_c(\mathbf{g}) = 0.8 \text{ GeV} \left(\frac{0.42}{|\mathbf{g}|} \right)^2. \quad (39)$$

9.3 First comparisons without pileup

These comparisons are exploratory tests of a few metrics and parameter choices, not the result of tuning. For the Cartesian configurations, I held the density radius at $d_c = 5$ cm and the density threshold at $\rho_c = 6$ GeV, and concentrated on the nearest-higher distance d_m . The preceding studies identified the distance reach as a particularly direct control of fragmentation, making it the first parameter to vary before a broader density-and-distance tuning. Unweighted and weighted Euclidean metrics were tested at $d_m = 3$ and 6 cm; Manhattan and weighted Chebyshev are shown at 6 cm. Default parameters are used for CLUE3D, which is the baseline.

The no-pileup comparison uses 20 000 generated single-photon events with energies of 10–250 GeV and $R = 0$ –350 cm in Run4D121 geometry, subject to the gun’s surface-acceptance requirements. Figure 23 shows that the 6 cm Cartesian configurations largely avoid the severe large-displacement loss of CLUE3D. The two Euclidean variants retain efficiencies in the

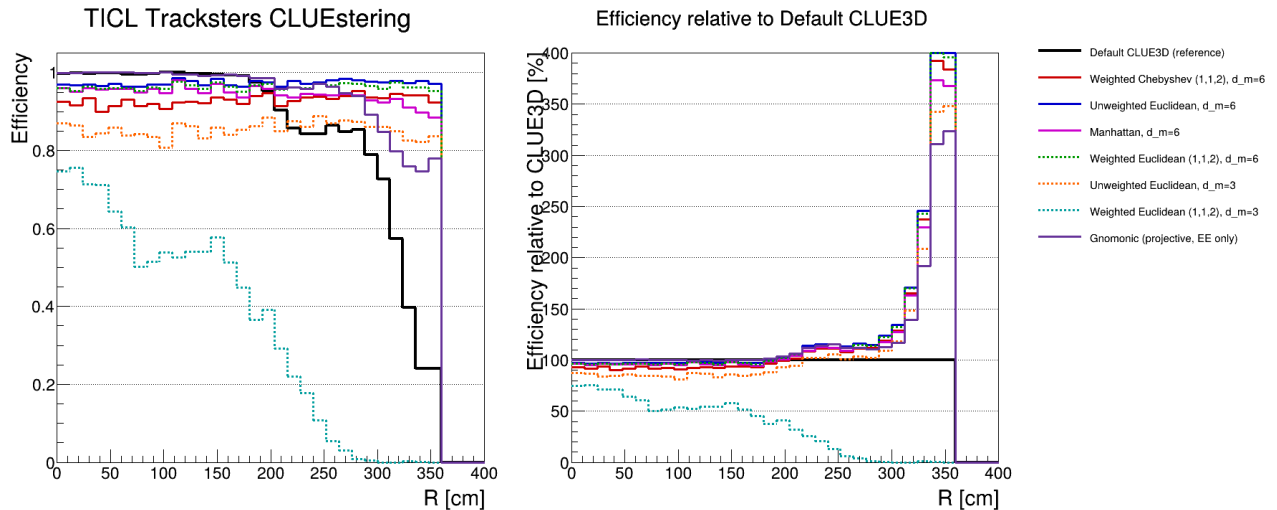


Figure 23: Single-photon efficiency without pileup for the exploratory CLUEsterng configurations, including the EE-only gnomonic configuration, with default CLUE3D as the baseline. The right panel shows $100 \epsilon / \epsilon_{\text{CLUE3D}}$; 100% denotes equal efficiency. The generated displacement scan ends at 350 cm.

mid-to-high 90% range over most of the scan; Manhattan and weighted Chebyshev also remain efficient, with somewhat lower curves. These simple choices already show that the default CLUE3D behaviour is not an unavoidable consequence of a displaced shower.

The gnomonic configuration performs well near the origin and degrades at large displacement, as expected for projective coordinates. Its decline is much less severe than that of CLUE3D, however, so this first test also looks promising. Sharing an origin projection does not imply identical performance: the cylinder neighbourhood, distance scales and density prescription differ from those of CLUE3D.

The surprise is the smaller Euclidean distance setting. Unweighted Euclidean at 3 cm is less efficient than at 6 cm, while the weighted 3 cm configuration loses almost all efficiency by approximately 300 cm. Since it uses Cartesian coordinates, this failure cannot be attributed to an origin projection.

9.4 Understanding the weighted Euclidean 3 cm failure

To understand why, I examined the nearest-higher links: their distance, their separate coordinate components, and the shower energy that ends up in the fragments they produce (Fig. 24). The links tell a simple geometric story. An inclined shower advances along a tilted axis, so consecutive layers are offset not only in z but also transversely. For a layer gap Δz , this transverse step is

$$\Delta r_{xy} \simeq |\Delta z| \tan \theta, \quad (40)$$

where θ is the photon polar angle. The HGCal layer spacing fixes the longitudinal part of each link, while a larger inclination—which is what a larger displacement produces—adds transverse distance. Panel (b) shows exactly this: at large R the transverse components grow while the longitudinal one barely changes. Once the transverse step outgrows the 3 cm reach, the

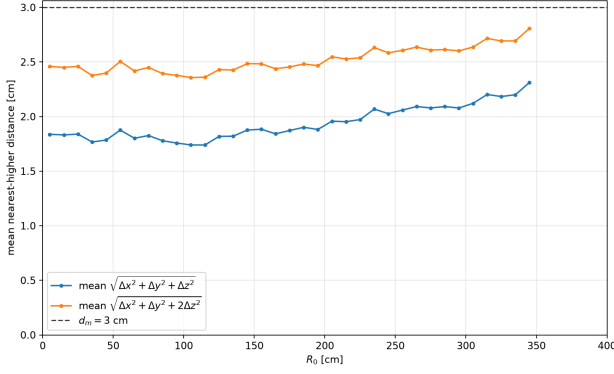
link is cut and a single continuous shower is split—even though nothing here projects towards the origin. The failure is therefore not really inherent to displaced particles, but to particles with large θ , which displaced particles simply tend to have.

One thing looks paradoxical at first: the mean weighted link distance stays below 3 cm across the whole scan (panel a), yet the efficiency collapses. The resolution is that clustering acts on individual links, and an average comfortably under the threshold can still hide many links above it. The start of the scan makes the point: below about 100 cm the mean photon angle and the mean link distance both fall, yet the most inclined fifth of the photons moves to larger θ (panel c). It is this growing angular tail, not the average shower, that runs into the distance limit.

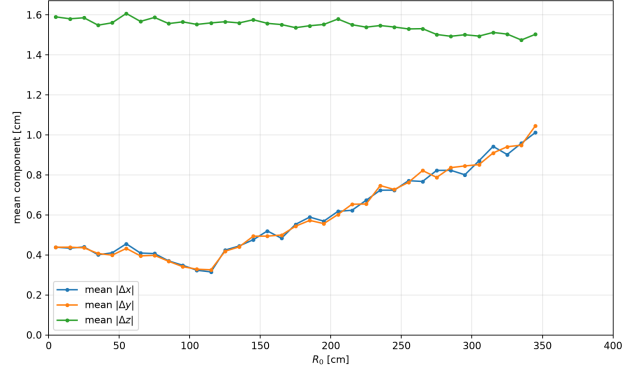
The fragmentation diagnostic ties this link-level failure to the lost energy (panel d). The energy in clusters that hang off a seed whose higher-density continuation lies beyond the reach rises to about two thirds of the total shower energy at large displacement, dominated by continuations just one layer away. A large part of each shower is therefore broken off into separate objects. The lesson of the 3 cm test is that Cartesian coordinates remove the artificial projective separation, but the distance reach and metric weights must still be large enough to follow an inclined shower from one layer to the next. This is a concrete target for future tuning.

9.5 Performance in pileup

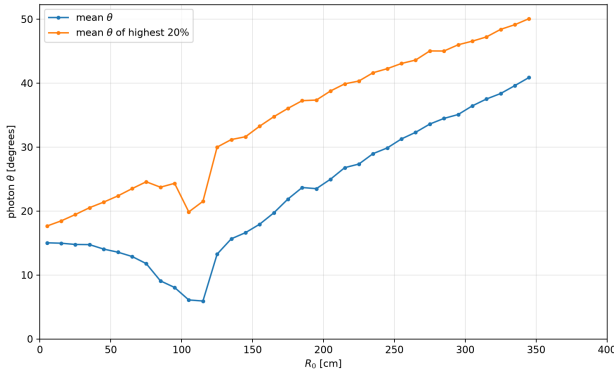
The PU200 comparison uses a 10 000-event sample for each tested configuration. As in Section 8, I measured the generated signal photon separately from the inclusive in-time truth population, and added two complementary rates: signal-photon duplicates and reconstructed-object merges. The signal measurement tests recovery of the displaced photon in a busy event; the inclusive one tests the response to the many addi-



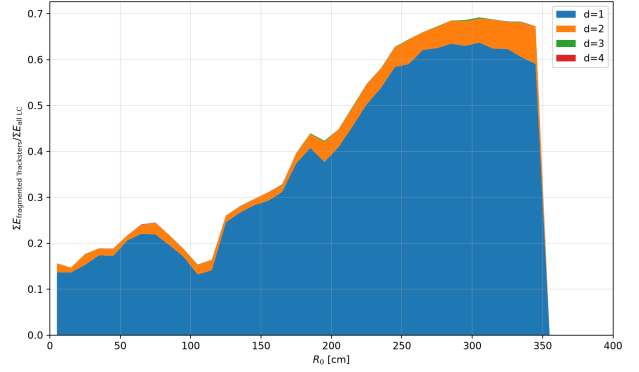
(a) Mean nearest-higher link distance, unweighted and weighted.



(b) Mean absolute components of the nearest-higher link.



(c) Mean photon angle θ , and the mean of its most-inclined fifth.



(d) Fraction of shower energy split into fragments, by the layer gap of the broken link.

Figure 24: Diagnostics for the weighted Euclidean metric at $d_m = 3$ cm, versus truth displacement R_0 . Panels (a) and (b) use only valid nearest-higher links. Panel (d) sums the full energy of the clusters rooted at seeds whose higher-density continuation lies beyond the 3 cm reach, normalized to the total LC energy; the label d in its legend is the physical layer gap of the broken link.

tional showers.

For the signal photon, the no-pileup ordering survives (Fig. 25a): the 6 cm Cartesian configurations stay efficient across the scan, the gnomonic configuration again beats CLUE3D at large displacement, and the weighted Euclidean 3 cm failure is still present. Pileup does not remove the geometric limitations, nor does it erase the gains of the more successful configurations.

The inclusive picture is different (Fig. 25b). The 6 cm Cartesian metrics remain competitive—unweighted Euclidean improves on CLUE3D over most of the range and weighted Chebyshev is broadly comparable—while Manhattan and the 3 cm choices recover much less. The gnomonic configuration is the striking outlier: even though its signal efficiency is the *higher* of the two, it collects only a small fraction of the inclusive population, several times less than CLUE3D does.

This is unlikely to be the projective metric itself. We assume its direction-dependent density threshold further suppresses much of the pileup, and it also hasn't been set up for hadronic subdetector which is relevant for PU. These are working explanations rather than conclusions, confirming them requires further checks.

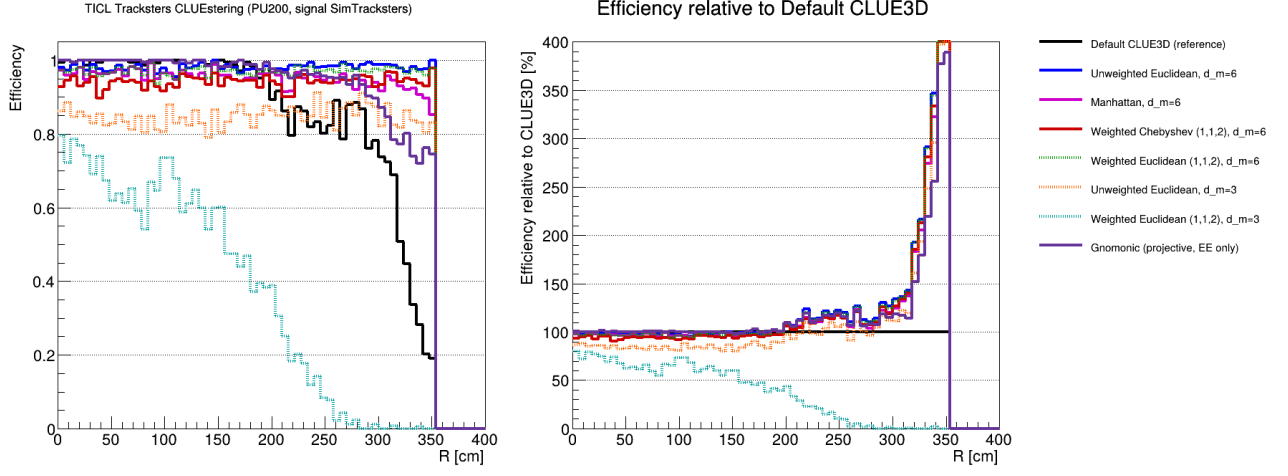
The duplicate rates in Fig. 26(a) show how completely the signal is collected into one object. The 6 cm Cartesian configurations remain close to zero, whereas

the 3 cm choices have appreciable duplicates at small and intermediate displacement. Gnomonic duplicates rise at large R , consistent with its remaining projective fragmentation; its efficiency gain over CLUE3D does not mean that every photon is recovered as one Trackster.

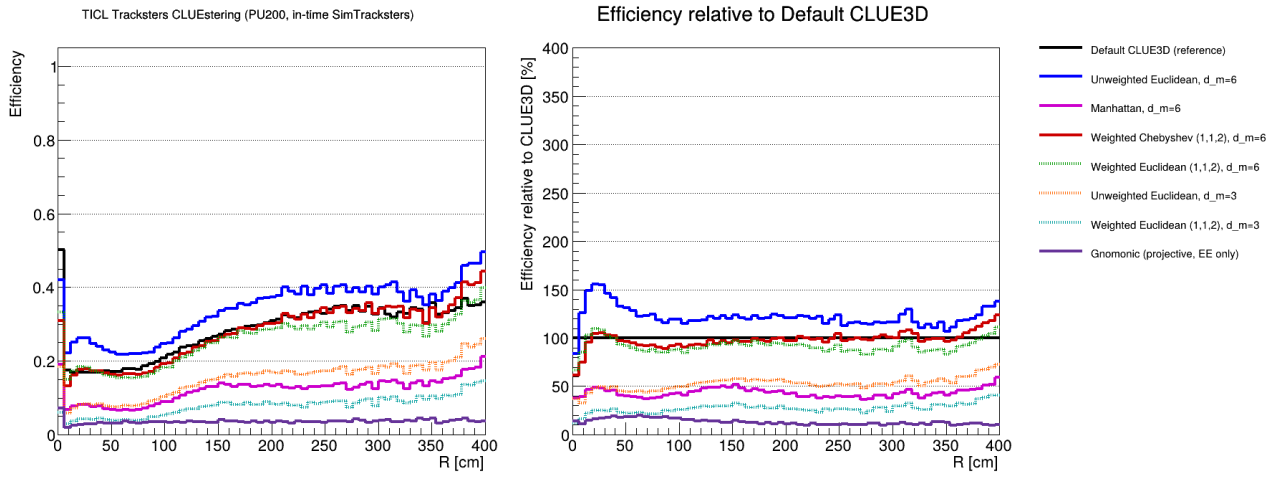
The reconstructed-object merge curves in Fig. 26(b) are small for the 6 cm Cartesian choices. Gnomonic and the 3 cm curves show more merging, with gnomonic broadly comparable in size to CLUE3D. These rates describe the surviving reconstructed objects, so their interpretation must account for the very different object populations and the efficiency loss. In particular, suppressing pileup objects can reduce opportunities to count merges without establishing better shower separation.

9.6 Two nearby photons and outlook

A dedicated two-photon comparison at 4 cm separation tests nearby showers directly (Fig. 27). Here efficiency cannot be read on its own: two overlapping photons collected into a single object still count as reconstructed, so a high efficiency can just as easily mean good recovery or outright merging. It has to be read together with the merge rate. On that basis the gnomonic configuration holds up well: at large displacement it keeps higher efficiency than CLUE3D

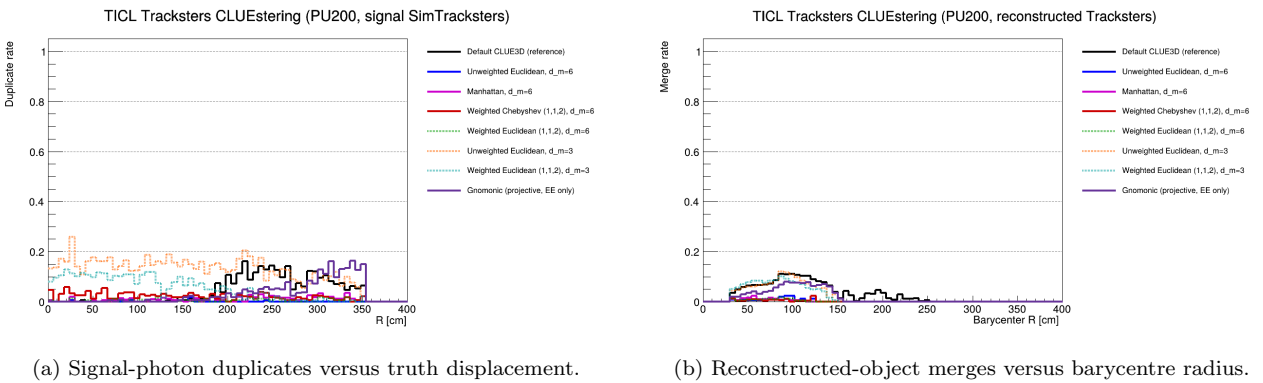


(a) Efficiency for the generated signal photon in PU200.



(b) Efficiency for the inclusive in-time truth population.

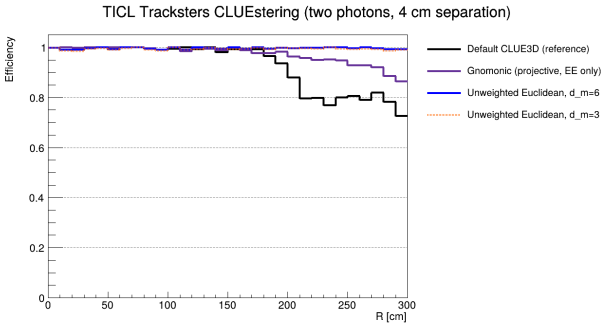
Figure 25: PU200 efficiencies, with default CLUE3D as the baseline and the same relative-efficiency convention as Fig. 23. The inclusive denominator includes showers in the hadronic sections, which the tested EE-only gnomonic configuration does not cluster.



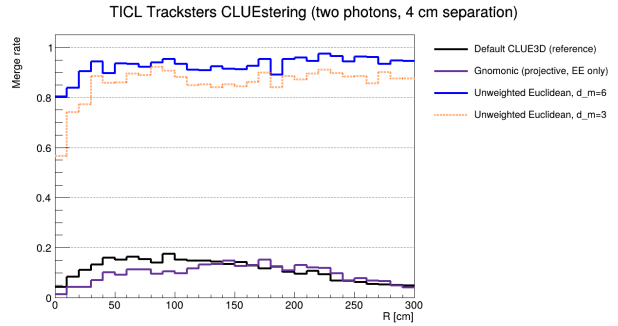
(a) Signal-photon duplicates versus truth displacement.

(b) Reconstructed-object merges versus barycentre radius.

Figure 26: Complementary PU200 rates. The duplicate denominator contains signal SimTracksters, while the merge denominator contains reconstructed Tracksters. The merge plot uses $\sqrt{x_c^2 + y_c^2}$, the transverse radius of the Trackster barycentre, rather than its back-extrapolated displacement at $z = 0$.



(a) Photon reconstruction efficiency.



(b) Reconstructed-object merge rate.

Figure 27: Two photons separated by 4 cm: photon efficiency and reconstructed-object merge rate for default CLUE3D, the gnomonic configuration, and unweighted Euclidean at $d_m = 6$ and 3 cm. Efficiency is binned in truth displacement and merge rate in reconstructed displacement, with 10 cm bins over 0–300 cm. The merge denominator contains nonempty Tracksters.

while its merge rate stays low and close to CLUE3D’s. It recovers more of each photon without merging the two together.

The Cartesian Euclidean metrics make the point in reverse. Unweighted Euclidean at $d_m = 6$ cm reaches near-perfect efficiency, but its merge rate is very high: the wide reach simply clusters both showers into one object, which is exactly what made it look so strong on single photons. More surprising is $d_m = 3$ cm. Its short reach fragmented a single shower earlier, yet with two photons its efficiency climbs back to near unity while it merges almost as heavily as the 6 cm case. The likely explanation is that the neighbouring photon bridges the gap: activity from the second shower fills the spacing that a 3 cm reach could not span alone, reconnecting the fragments into one merged object. This would need a dedicated study to confirm.

As shown in Section 7, fragmentation can also hide mixing inside an object-based merge rate, so extending the cross-shower LC-link diagnostic to these configurations would test separation more directly. Taken together, the results show that CLUEstering offers several useful choices: the gnomonic configuration improves on CLUE3D in both single- and two-photon tests, while the Cartesian metrics buy their high efficiency partly by merging and must be tuned against that. Its strong suppression of the pileup remains a limitation of the present setup. The next step is to extend the input coverage and tune coordinates, metric weights, distance reaches and density settings together, using signal and PU efficiency, duplicate, merge and fake rate to guide the choice.

10 Conclusions

This project looked at how HGCal clustering behaves when photons no longer point back to the interaction point, and which specific reconstruction features cost performance. Along the way I improved the displaced-particle gun, added displacement and timing observables to the CMSSW validation, made the association thresholds configurable, and produced the two-dimensional performance maps used throughout.

Comparing shower-axis estimators gave an approximately 10 cm resolution on the back-extrapolated displacement of the most energetic Trackster. Small fragments do much worse than this, so it should not be read as a number that describes every reconstructed object.

A large part of the work went into explaining why default CLUE3D loses single displaced photons. Because it projects everything from the origin, two points on the same shower axis look transversely separated by roughly $R|\Delta z|/z$, and this apparent separation grows with displacement. The loss comes in two stages: around $R = 200$ cm the energy carried by fragments whose nearest-higher link skips two layers rises and then levels off, which produces the intermediate plateau, and at larger displacement one-layer breaks take over and drive the final decline. Widening the seed distance from 1.8 cm to 2.1 cm, with the density radius fixed, pushes the fragmentation out to larger displacement and recovers substantially more of the shower.

The two-photon studies make the case that energy recovery cannot be judged on its own. For photons 4 cm apart, the energy linked across the two showers keeps rising with displacement even where the object-based merge rate is falling, so fragmentation can make the merge rate look better while the mixing underneath is getting worse. In PU200 the injected signal photon behaves much as it does without pileup. The inclusive truth population is a different story: its particle mix, kinematics and shower sizes all shift with resolved displacement, and charged particles bend in the magnetic field, so its displacement curve cannot be read as a clean scan of production radius.

Comparing metrics through CLUEstering showed both promise and limits. Several Cartesian configurations stay efficient on single displaced photons across the whole scan, including in pileup, but the Euclidean ones merge the 4 cm photon pair heavily, so high efficiency alone does not mean the two particles were actually resolved. The gnomonic configuration was the most encouraging, giving higher efficiency than default CLUE3D at large displacement while keeping a low merge rate on the two-photon test, though this still needs further studies and its EE-only coverage is why

its inclusive pileup efficiency looks so low.

The natural next step is to tune coordinates, metric weights, distance scales and density thresholds together, judging them against energy recovery, fragmentation and cross-shower mixing at once. It also needs to be tested on more realistic inputs: the controlled photons here are injected just before HGCAL, so they say nothing about upstream interactions or a real LLP lifetime. The generation and validation tools built for this study give a starting point for that work, and the geometric picture points to the specific clustering decisions worth changing.

Acknowledgements

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